

The Geometry of Informativeness: A Bregman-Centroid Diagnostic for Partial Pooling in Multisite Trials

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ABSTRACT. Multisite randomized trials are increasingly analyzed for site-specific effects: how large the effect was at each site, which sites benefited most, and how the effects are distributed. Recent work has shown that the attainable quality of all three answers is well summarized by a single index, the informativeness of the trial, which compares the between-site variation in true effects with a geometric-mean summary of the within-site noise. Two questions have remained open: why a geometric mean appears, and what a single average conceals. This paper answers both with one geometric observation. The weight that determines how far each site’s estimate is pulled toward the overall average is a reliability coefficient in a projection sense, and informativeness is exactly the centroid of the site reliabilities under the geometry natural for quantities between zero and one; averaging on that geometry’s own scale is where the geometric mean comes from. Because a centroid is a projection, it leaves a residual, and the residual measures how unevenly reliability is spread across sites. We prove that this dispersion is linked exactly to the gap between informativeness and the average reliability that sets benchmark estimation risk, with a direction governed by the shape of the reliability profile, and a Monte Carlo experiment shows that the profile predicts estimator performance where the index alone cannot. The same geometry yields a balanced-allocation design rule, and the usual plug-in estimate of informativeness is shown to be systematically optimistic. The Online Credit Recovery Study illustrates the workflow.

Keywords: multisite trials; site-specific treatment effects; shrinkage; reliability; informativeness; Bregman divergence

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1. Introduction

Multisite randomized trials are widely used in education research because they can answer several questions at once (Raudenbush & Bloom, 2015; Weiss et al., 2017). Beyond the average treatment effect, a district may want to know how well the intervention worked in its own schools, a developer which sites struggled, and a policymaker how much impacts vary across contexts. These aims translate into three inferential goals for the site-specific effects: estimating the individual effects, ranking them, and estimating their distribution (Lee et al., 2025; Shen & Louis, 1998). The difficulty is the same for all three goals. Each site provides only a noisy estimate of its own effect, so any analysis must borrow strength across sites, and how much the borrowing helps depends on how the within-site noise compares with the between-site signal.

Lee et al. (2025) showed in a large simulation study that this comparison can be summarized by a single index they call the informativeness of the trial, denoted I . The index contrasts the variation in true effects across sites with a geometric-mean summary of the within-site sampling variances, and in their study it was the most influential factor governing which combination of prior model and posterior summary performs best for each inferential goal. When I is low, flexible semiparametric machinery adds little and targeted posterior summaries matter most; when I is high, flexible models recover the effect distribution well. The index has a direct interpretation as the average reliability of the site-specific estimates, and it can be computed at the outset of an analysis.

Two questions about this index have remained open. The first is why the geometric mean appears in it. Classical heterogeneity summaries in meta-analysis, such as the I^2 statistic, rest on arithmetic averaging of within-study variances (Higgins & Thompson, 2002), and Lee et al. (2025) offer no derivation that singles out the geometric mean; the formula is justified by its empirical performance. The second question is what a single average conceals. Two trials can share the same I while one spreads its reliability evenly across sites and the other mixes a handful of precise sites with many uninformative ones. If such profiles behave differently, an analyst who sees only the average has no way to know, and as far as we are aware no diagnostic in the multisite literature addresses this.

In this paper we address both questions with a single geometric observation. The site-level shrinkage weight, the quantity that determines how far each site’s estimate is pulled toward the grand mean, is exactly a reliability coefficient in the projection sense

recently formalized by Zumbo (2026a): the proportion of variance in the observed site estimate that is attributable to the true effect, with no reference to hypothetical repeated testing. A multisite trial therefore carries one reliability per site, and the informativeness of Lee et al. (2025) is exactly the centroid of these site reliabilities under the Kullback–Leibler geometry that is natural for quantities bounded between zero and one (Banerjee et al., 2005; Nielsen & Nock, 2009; Zumbo, 2026b). Centroids in this geometry average the reliabilities on the scale of log signal-to-noise ratios, and averaging on that scale is algebraically the same as taking the geometric mean of the sampling variances. Once this way of averaging is adopted, the geometric mean follows. There are also reasons to adopt it: the underlying divergence is the canonical one for quantities on the unit interval, and it is the only member of the standard centroid family that reproduces the index education researchers already use.

The identification does useful work beyond settling the derivation. Because a centroid is a projection, it leaves a residual, and the residual is itself a diagnostic: it measures how much the reliability profile disperses around the reported index, and a Pythagorean decomposition shows that it is the irreducible lack of fit of any single-number summary of the profile. The dispersion also carries a risk interpretation. Benchmark estimation risk for the site effects is set by the arithmetic average of the site reliabilities rather than by the index itself, and an exact identity ties the gap between these two averages to the rate at which the dispersion changes along the informativeness scale. For nearly symmetric profiles the gap is positive below the midpoint of the reliability scale and negative above it, so modest dispersion lowers benchmark risk in weakly informative trials and raises it in strongly informative ones; the profile’s skew can overturn this local rule, and our simulations therefore examine the dispersion effect separately for each profile shape. A companion quantity, an effective number of informative sites, records whether the reliability mass is carried broadly or by a few dominant sites.

The same geometry has a design consequence and an estimation consequence. For a fixed total sample size, balanced allocation across sites maximizes informativeness, and the loss from imbalance is governed by the ratio of the geometric to the arithmetic mean site size; this explains why Lee et al. (2025) found the coefficient of variation of site sizes to be a second-order design factor. On the estimation side, the usual plug-in computation systematically overstates informativeness, because informativeness is concave in the between-site variance, and we recommend computing all of the diagnostics as posterior functionals, with credible intervals, instead.

Some remarks on the scope of the paper are in order. We propose no new estimator of site-specific effects; the diagnostics inform the choice among existing tools. We do not claim that the identity behind the reliability interpretation of the shrinkage weight is technically deep; it is a variance decomposition, and its usefulness lies in connecting multisite analysis with psychometric reliability theory and with the Bregman machinery that produces the new diagnostics. We do not offer universal numeric thresholds for the diagnostics, because their consequences depend on the inferential goal, the number of sites, and the loss of interest; our simulations map those consequences instead. Finally, our empirical results are confined to the normal-normal working model with the posterior summaries studied by Lee et al. (2025), because that is the setting in which I was proposed and is used.

The plan of the paper is as follows. Section 2 reviews the multisite model, the three inferential goals, and the informativeness index. Section 3 establishes the reliability interpretation of the shrinkage weights. Section 4 states the centroid characterization and discusses what does, and does not, single out the geometric mean. Section 5 introduces the dispersion and concentration diagnostics, their risk interpretation, and their estimation. Section 6 develops the design result. Section 7 reports a simulation study of what the new diagnostics add empirically, and Section 8 applies the methodology to the Online Credit Recovery Study. Section 9 concludes. All proofs, along with extensions and additional results, appear in the appendices.

2. Site-Specific Inference and the Informativeness Index

2.1. The Multisite Model. We work in the meta-analytic formulation of a multisite trial introduced by Rubin (1981). A trial comprises J sites. Within site j , randomization yields an estimate $\hat{\tau}_j$ of the true site-average treatment effect τ_j , together with a squared standard error $v_j = \widehat{se}_j^2$ computed from the site's own data and treated as known. The two-level model is

$$\hat{\tau}_j \mid \tau_j \sim \mathbf{N}(\tau_j, v_j), \quad \tau_j \mid \tau, \sigma^2 \sim \mathbf{N}(\tau, \sigma^2), \quad j = 1, \dots, J, \quad (1)$$

where τ is the average effect across the population of sites and σ^2 is the cross-site variance of true effects. Under simple randomization with site sample size n_j , treated proportion p_j , common within-arm outcome variance s_j^2 , and unit-level covariates explaining a proportion R^2 of outcome variance, the sampling variance follows the

standard Neyman form

$$v_j = \left(\frac{1}{p_j} + \frac{1}{1 - p_j} \right) \frac{s_j^2(1 - R^2)}{n_j}, \quad (2)$$

so that better-staffed sites, balanced allocation, and prognostic covariates all act through v_j (Lee et al., 2025). Our estimands are finite-population: the collection $\{\tau_j\}_{j=1}^J$ for the sites actually in the trial, their ranks, and their empirical distribution function (Lee et al., 2025; Shen & Louis, 1998).

Conditional on the hyperparameters, the posterior for each site effect is normal,

$$\tau_j \mid \tau, \sigma^2, \hat{\tau}_j \sim \mathcal{N}(\tau_j^*, \sigma^2(1 - S_j)), \quad \tau_j^* = \tau + (\hat{\tau}_j - \tau) S_j, \quad (3)$$

with shrinkage-to-data weight

$$S_j = \frac{\sigma^2}{\sigma^2 + v_j} \in (0, 1). \quad (4)$$

The weight S_j measures how much of the observed site estimate is retained after pooling: a site with $v_j \ll \sigma^2$ keeps nearly all of its own estimate, and a site with $v_j \gg \sigma^2$ is pulled almost entirely to the grand mean. The complementary pooling weight $\omega_j = 1 - S_j$ is, under the random-coefficient reading of model (1), the site-level pooling factor of Gelman and Pardoe (2006).

2.2. Inferential Goals, Losses, and Posterior Summaries. Following Shen and Louis (1998) and Lee et al. (2025), we evaluate an estimate vector $a = (a_1, \dots, a_J)$ of the site effects against three losses: the mean squared error loss $\text{MSEL} = J^{-1} \sum_j (a_j - \tau_j)^2$, reported as its square root RMSE; the mean squared error loss of percentiles, $\text{MSELP} = J^{-1} \sum_j (A_j/J - R_j/J)^2$, where R_j and A_j are the true and estimated ranks; and the integrated squared error loss for the empirical distribution function (EDF), $\text{ISEL} = \int \{A(t) - G_J(t)\}^2 dt$, where G_J is the empirical distribution of the true τ_j and A that of the estimates. Different posterior summaries target different losses. The posterior means (PM) τ_j^* minimize expected MSEL but are underdispersed as an ensemble; the constrained Bayes summary (CB; Ghosh, 1992; Louis, 1984) rescales the posterior means so their finite-population variance matches the estimated marginal variance of the τ_j ; and the triple-goal summary (GR; Shen & Louis, 1998) reads off evenly spaced quantiles of the estimated effect distribution in the order of the posterior expected ranks. Because shrinkage weights differ across sites whenever the v_j differ, the posterior means are not a monotone transformation of the raw estimates, so even the ranking of sites depends on the reliability profile; we return to this in

Section 7. Producing a ranking is weaker than quantifying uncertainty about ranks, for which compound-decision and rank-confidence-set methods exist (Gu & Koenker, 2023; Mogstad et al., 2024); the diagnostics below indicate whether rank-directed summaries are worth computing, not whether any particular rank is significant.

2.3. The Informativeness Index and Related Quantities. To summarize how much the trial as a whole can say about its site effects, Lee et al. (2025) define the informativeness

$$I = \frac{\sigma^2}{\sigma^2 + \tilde{v}}, \quad \tilde{v} = \exp\left(\frac{1}{J} \sum_{j=1}^J \ln v_j\right), \quad (5)$$

the shrinkage weight that a hypothetical site would have if its sampling variance equaled the geometric mean $\tilde{v}$ of the observed sampling variances. In their simulations, spanning 1,500 conditions calibrated to education multisite trials, I ranged from .01 to .71 and organized the performance of nine model-summary combinations across all three losses, and it is the natural quantity to compute first. Lee et al. (2025) describe I as the average reliability of the site-specific estimates. This description raises the two questions of this paper: in what sense each S_j is a reliability, and in what sense I is their average.

Table 1 relates these quantities to established summaries in meta-analysis and multilevel modeling. The I^2 statistic of Higgins and Thompson (2002) measures the proportion of total variation in the observed effect estimates attributable to heterogeneity, using an arithmetic-type pooling of within-study variances, and answers a different question: how much heterogeneity is present, rather than how well site-specific parameters can be recovered. The pooling factor of Gelman and Pardoe (2006) is a site-level quantity equal to $1 - S_j$ in our setting. The generalizability index of Tipton (2014) is also a scalar summary of what a trial can support, but of a third kind again: it measures how well the experimental sample represents an inference population, whereas I measures how recoverable the sampled sites' own effects are. Closer to the aggregation question considered here, psychometrics has long combined reliability coefficients across studies and debated the appropriate averaging transformation, from Fisher-type transformations to raw means (Feldt & Charter, 2006), and the reliability-generalization literature compares such pooling strategies systematically (López-Ibáñez et al., 2024). That work aggregates coefficients across instruments and studies rather than within one trial's shrinkage profile, and as far as we are aware none of it connects an averaging rule to a Bregman geometry or to downstream estimator performance. The informativeness I is a trial-level summary

TABLE 1. Site-level and trial-level summaries of a multisite reliability profile, and related quantities.

Quantity	Definition	Level and scale	Question answered
Shrinkage weight S_j	$\sigma^2/(\sigma^2 + v_j)$	Site; $(0, 1)$	How reliable is site j 's estimate? (Proposition 1: a projection reliability, $\cos^2 \theta_j$)
Pooling factor ω_j	$1 - S_j$ (Gelman & Par-doe, 2006)	Site; $(0, 1)$	How far is site j pooled toward the mean? ($\sin^2 \theta_j$)
Informativeness I	$\sigma^2/(\sigma^2 + \tilde{v})$, $\tilde{v} = \text{GM}(v_j)$ (Lee et al., 2025)	Trial; $(0, 1)$	Where is the reliability profile centered? (Theorem 1: left KL centroid)
Mean reliability $\bar{S}$	$J^{-1} \sum_j S_j$ (right centroid)	Trial; $(0, 1)$	What benchmark MSEL risk is achievable? (Proposition 2: risk = $\sigma^2(1 - \bar{S})$)
Dispersion H	$J^{-1} \sum_j \text{KL}\{\text{Bern}(I) \parallel \text{Bern}(S_j)\}$	Trial; $[0, \infty)$	How badly does any single summary fit the profile? (Theorem 2)
Effective sites J_{eff}	$(\sum_j S_j)^2 / \sum_j S_j^2$	Trial; $[1, J]$	How many sites carry the reliability mass?
Heterogeneity I^2	share of total variation due to heterogeneity (Higgins & Thompson, 2002)	Trial; $[0, 1)$	Is there cross-site variation to study at all?

Note. v_j denotes the squared standard error of the site- j estimate, σ^2 the cross-site variance of true effects, and GM the geometric mean. I^2 conditions on observed estimates and targets the presence of heterogeneity; S_j , I , H , and J_{eff} target the recoverability of site-specific parameters. KL = Kullback–Leibler.

of the S_j , and the diagnostics introduced below, H and J_{eff} , describe the shape of the reliability profile that I reduces to a single number.

3. Shrinkage Weights as Reliability Coefficients

3.1. Reliability as Projection Efficiency. In classical test theory, reliability is defined with reference to hypothetical repeated administrations of a test to the same examinee. Zumbo (2026a) argues that no such replications are needed and defines reliability directly: given an observed score X and the information $\mathcal{G}$ that identifies the object being measured, the true score is the conditional expectation $T = \mathbb{E}[X \mid \mathcal{G}]$,

and reliability is the variance ratio

$$\text{Rel}(X) = \frac{\text{Var}(T)}{\text{Var}(X)} = \frac{\text{Var}(\mathbb{E}[X \mid \mathcal{G}])}{\text{Var}(X)}. \quad (6)$$

Because conditional expectation is an orthogonal projection in the space of square-integrable random variables, the decomposition $X = T + E$ with T uncorrelated with $E = X - T$ holds automatically, and $\text{Rel}(X)$ equals the squared cosine of the angle between the centered X and its projection T , a measure of projection efficiency (Zumbo, 2026a). The definition makes no reference to test forms or repeated administration; it requires an observed quantity, a target of measurement, and a joint distribution.

A multisite trial supplies all three: the observed quantity is the site estimate $\hat{\tau}_j$, the target is the site's true effect τ_j , and the joint distribution is given by model (1). In generalizability-theory terms, the trial is a one-facet measurement design in which sites are the objects of measurement and the within-site experiment is the measurement procedure, so the ratio of universe-score variance to observed-score variance is precisely of the form (6) (Brennan, 2001; Cronbach et al., 1972). Taking $X_j = \hat{\tau}_j$ and $\mathcal{G}_j = \sigma(\tau_j)$, the conditional expectation is $\mathbb{E}[\hat{\tau}_j \mid \tau_j] = \tau_j$, and the variance decomposition gives the following identity.

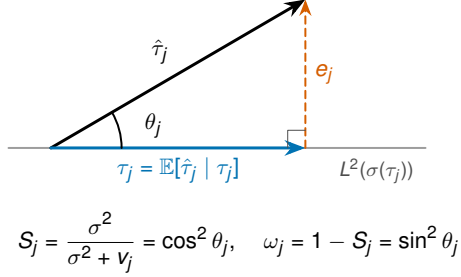
Proposition 1 (Shrinkage weights are projection reliabilities). Under model (1), the reliability of the site estimate $\hat{\tau}_j$ for its true effect τ_j equals the shrinkage weight:

$$\text{Rel}(\hat{\tau}_j) = \frac{\text{Var}(\tau_j)}{\text{Var}(\hat{\tau}_j)} = \frac{\sigma^2}{\sigma^2 + v_j} = S_j = \cos^2 \theta_j,$$

where θ_j is the angle between the centered $\hat{\tau}_j$ and τ_j in L^2 . Equivalently, the pooling factor satisfies $\omega_j = 1 - S_j = \sin^2 \theta_j$. Only the first-stage conditional moment conditions, $\mathbb{E}[\hat{\tau}_j \mid \tau_j] = \tau_j$ and $\mathbb{E}\{\text{Var}(\hat{\tau}_j \mid \tau_j)\} = v_j$, together with a finite positive σ^2 , are needed; normality plays no role. We prove this, and collect the boundary cases $v_j \rightarrow 0$ and $\sigma^2 \rightarrow 0$, in Appendix A.

The identity itself is elementary, and describing S_j as the reliability of $\hat{\tau}_j$ is textbook usage (Rabe-Hesketh & Skrondal, 2022). What Proposition 1 adds is the projection interpretation, illustrated in panel (a) of Figure 1. Each site of a multisite trial carries a well-defined measurement-quality coefficient on the common scale $(0, 1)$, comparable across sites and equal to the fraction of observed variance that is signal, and the pooling factor is its complement: shrinkage and pooling are the squared cosine and squared sine of the same angle, so the identity $S_j + \omega_j = 1$ expresses

(a) Site level: reliability as projection



(b) Trial level: informativeness as centroid

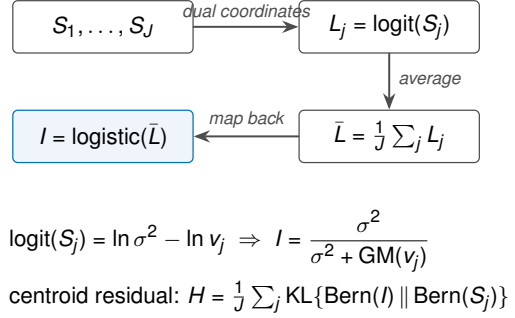


FIGURE 1. The two-level geometry of a multisite reliability profile. Panel (a): within each site, the estimate $\hat{\tau}_j$ decomposes orthogonally into the true effect τ_j and sampling error e_j ; the shrinkage weight is the squared cosine of the projection angle (Proposition 1). Panel (b): across sites, the informativeness I of Lee et al. (2025) averages the site reliabilities in their dual (logit) coordinates, which is exactly the left Bregman centroid under the Bernoulli–Kullback–Leibler geometry (Theorem 1); the residual of this projection is the dispersion diagnostic H of Section 5.

orthogonality. The same ratio is also a squared correlation, a regression R^2 , and a communality (Zumbo, 2026a), so tools developed for reliability coefficients transfer to shrinkage weights. In the next section we consider one such tool, an averaging rule.

4. The Informativeness Index as a Kullback–Leibler Centroid

4.1. The Aggregation Problem. We now consider how J reliabilities should be combined into one number. The arithmetic mean $\bar{S} = J^{-1} \sum_j S_j$ is the obvious candidate, but reliabilities are confined to $(0, 1)$, where arithmetic averaging behaves poorly near the boundaries and many equally natural means disagree. Instead of choosing among means by convention, we treat the question geometrically. A mean of $S_1, \dots, S_J$ is a point p that is closest to the profile on average, and closeness requires a divergence. For a strictly convex, differentiable potential ϕ on $(0, 1)$, the Bregman divergence

$$D_\phi(x, y) = \phi(x) - \phi(y) - \phi'(y)(x - y) \geq 0 \quad (7)$$

measures the gap between ϕ at x and its tangent-line approximation at y (Bregman, 1967). Each choice of ϕ induces a centroid, the minimizer of the average divergence to the profile. Centroids in the first argument, called left centroids, have a closed form:

they average in the dual coordinates ϕ' and map back, $I_\phi = (\phi')^{-1}\{J^{-1}\sum_j \phi'(S_j)\}$ (Banerjee et al., 2005; Nielsen & Nock, 2009). Squared Euclidean distance, $\phi(x) = x^2/2$, has identity dual coordinates, and its centroid is the arithmetic mean. The geometry adapted to probabilities is different: taking ϕ to be the negative Bernoulli entropy, $\phi_{\text{KL}}(p) = p \ln p + (1-p) \ln(1-p)$, the induced divergence is the Kullback–Leibler divergence between Bernoulli distributions, $D_{\phi_{\text{KL}}}(p, q) = \text{KL}\{\text{Bern}(p) \parallel \text{Bern}(q)\}$, and the dual coordinate is the logit (Zumbo, 2026b). In this geometry, averaging happens on the logit scale.

Theorem 1 (Centroid characterization of informativeness). Let $S_1, \dots, S_J \in (0, 1)$ be the site reliabilities of model (1). The informativeness (5) of Lee et al. (2025) is the left Bregman centroid of the reliability profile under the Bernoulli–Kullback–Leibler geometry:

$$I = \arg \min_{p \in (0,1)} \frac{1}{J} \sum_{j=1}^J \text{KL}\{\text{Bern}(p) \parallel \text{Bern}(S_j)\},$$

the minimizer exists, is unique, and satisfies the dual-coordinate identity

$$\text{logit}(I) = \frac{1}{J} \sum_{j=1}^J \text{logit}(S_j).$$

The proof is short and appears in Appendix B: the first-order condition equates $\phi'(p)$ with the average of the $\phi'(S_j)$, and $\text{logit}(S_j) = \ln \sigma^2 - \ln v_j$, so averaging logits averages $\ln v_j$, and applying the inverse logit returns expression (5) with the geometric mean $\tilde{v}$.

Theorem 1 answers the first open question to the extent that a characterization can. The inverse-gradient step it rests on is classical convex analysis (Banerjee et al., 2005; Nielsen & Nock, 2009); what is new is the identification: the index that Lee et al. (2025) arrived at empirically is the left Bernoulli–KL centroid of the site reliabilities, so its geometric mean is a consequence of logit-scale averaging rather than an arbitrary smoothing choice. The geometry itself, however, remains a choice. A different potential ϕ , or the right-centroid orientation, which by the same classical result yields the arithmetic mean $\bar{S}$ (Appendix B), produces a different and equally coherent average, and Table 2 collects three members of the centroid family that bracket I in practice: the arithmetic, KL, and harmonic centroids.

Nothing below requires the family in general form. The KL member is singled out because it coincides with the index of Lee et al. (2025), whose behavior we set out to

TABLE 2. Left Bregman centroids of the reliability profile $S_1, \dots, S_J$ under three potentials.

Potential $\phi(x)$	Dual coordinate $\phi'(x)$	Centroid I_ϕ	Interpretation
$\frac{1}{2}x^2$ (squared Euclidean)	x	$\bar{S} = \frac{1}{J} \sum_j S_j$	Arithmetic mean (right centroid); governs benchmark MSEL risk (Proposition 2)
$x \ln x + (1-x) \ln(1-x)$ (Bernoulli entropy)	$\text{logit}(x)$	$\text{logistic}(\frac{1}{J} \sum_j \text{logit } S_j)$	Kullback–Leibler centroid; equals the informativeness I of Lee et al. (2025) (Theorem 1)
$-\ln x$ (Burg entropy)	$-1/x$	$(\frac{1}{J} \sum_j S_j^{-1})^{-1}$	Harmonic mean; dominated by the least reliable sites

Note. Each left centroid averages the profile in its dual coordinates, $I_\phi = (\phi')^{-1}\{J^{-1} \sum_j \phi'(S_j)\}$ (Banerjee et al., 2005; Nielsen & Nock, 2009). The three centroids coincide if and only if the profile is constant, and their disagreement grows with the dispersion H ; in particular, the arithmetic-KL gap satisfies $\bar{S} - I = \partial H / \partial \text{logit}(I)$ exactly and $\approx (1 - 2I)H$ for near-symmetric profiles (Proposition 2), so the left and right centroids bracket the profile and their gap is the risk-relevant expression of its dispersion.

explain, and because Bernoulli–KL is the canonical divergence for probability-scale quantities. What the identification provides is developed in the next two sections: a residual, an exact link between that residual and estimation risk, and a design result. When the profile is homogeneous all centroids coincide, and their disagreement grows with the dispersion of the profile, to which we now turn.

5. Dispersion and Concentration Diagnostics

5.1. A Pythagorean Decomposition and the Dispersion Diagnostic. A centroid is a projection of the profile $(S_1, \dots, S_J)$ onto the one-dimensional manifold of constant profiles $\{(p, \dots, p)\}$, and the projection leaves a residual. Figure 2 illustrates why the residual matters: the three trials shown share the same informativeness $I = .30$ exactly, yet one has a flat reliability profile, one spreads its reliability evenly across two orders of magnitude of sampling variance, and one concentrates its reliability in a handful of precise sites. Define the *Bregman dispersion* of the reliability profile as the average divergence from the centroid to the sites,

$$H = \frac{1}{J} \sum_{j=1}^J \text{KL}\{\text{Bern}(I) \parallel \text{Bern}(S_j)\} \geq 0, \quad (8)$$

with $H = 0$ if and only if all sites share one reliability. The next result locates H within the selected geometry: under the Bernoulli–KL potential, H is the exact

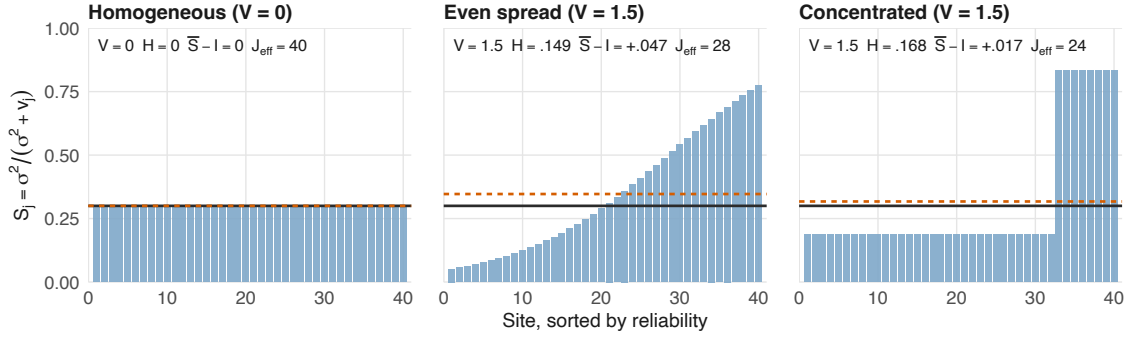


FIGURE 2. Three reliability profiles with identical informativeness. Each panel shows the sorted site reliabilities S_j of a $J = 40$ trial with $I = .30$ exactly; the solid line marks the common centroid I and the dashed line the arithmetic mean $\bar{S}$, whose gap $\bar{S} - I$ is the risk-relevant expression of the dispersion (Section 5.2). The printed values give the logit-scale variance V , the dispersion H , the gap $\bar{S} - I$, and the effective number of informative sites J_{eff} for each profile.

residual that remains when any constant summary of the profile is assessed by its average divergence. A different potential or orientation would define a different, equally coherent residual.

Theorem 2 (Constant-manifold Pythagorean decomposition). For every $p \in (0, 1)$,

$$\frac{1}{J} \sum_{j=1}^J D_\phi(p, S_j) = D_\phi(p, I_\phi) + H_\phi, \quad H_\phi = \frac{1}{J} \sum_{j=1}^J D_\phi(I_\phi, S_j),$$

for any strictly convex, differentiable potential ϕ , where I_ϕ is the left centroid; conversely, if the display holds for all p with some center c in place of I_ϕ , then $c = I_\phi$. Under ϕ_{KL} this yields decomposition of the average KL lack-of-fit of any candidate summary p into a location term $\text{KL}\{\text{Bern}(p) \parallel \text{Bern}(I)\}$ and the irreducible dispersion H of definition (8). Proof in Appendix C.

Theorem 2 is the analogue, in the reliability geometry, of the decomposition of a sum of squares around a mean: total equals location plus spread, and the KL centroid plays the role of the mean because it is the unique center for which the cross term vanishes for every reference point. In this geometry, H is the amount by which any single number, including I itself, misdescribes the reliability profile on average. When H is near zero the profile is nearly constant, and reporting I alone loses almost nothing. When H is large, no single number describes the profile well, and quantities

that depend on the profile through more than its centroid, which Section 5.2 shows includes estimation risk, can differ from what I suggests.

5.2. Dispersion and Benchmark Estimation Risk. The relevance of H to an analyst who already knows I lies in the fact that different inferential objects average the reliability profile in different ways, and the gap between the two most important averages, the KL centroid I and the arithmetic mean $\bar{S}$, is a derivative of H . Write $V = J^{-1} \sum_j \{\text{logit}(S_j) - \text{logit}(I)\}^2$ for the variance of the profile on the logit scale; because $\text{logit}(S_j) = \ln \sigma^2 - \ln v_j$, the quantity V equals the variance of the $\ln v_j$ and is conditionally fixed given the first-stage variance estimates, requiring no propagation through the posterior of σ^2 (though it does inherit whatever estimation error the v_j themselves carry; Section 8 quantifies this).

Proposition 2 (Dispersion and the centroid-risk gap). Write $m = \text{logit}(I)$ and hold the centered logit profile $\Delta_j = \text{logit}(S_j) - m$ fixed, so that $J^{-1} \sum_j \Delta_j = 0$ and $J^{-1} \sum_j \Delta_j^2 = V$; let $\rho_3 = J^{-1} \sum_j |\Delta_j|^3$. Then:

$$\begin{aligned} \text{(i)} \quad H(m; \Delta) &= \frac{1}{J} \sum_{j=1}^J \ln(1 + e^{m+\Delta_j}) - \ln(1 + e^m) \quad \text{and} \quad \frac{\partial H}{\partial m} = \bar{S} - I \quad \text{exactly;} \\ \text{(ii)} \quad H &= \frac{1}{2} I(1 - I) V + O(\rho_3) \quad \text{and} \quad \bar{S} - I = (1 - 2I) H + O(\rho_3); \end{aligned}$$

and (iii) with known hyperparameters the average posterior variance of the site effects, which is the expected MSEL of the known-hyperparameter Gaussian linear-shrinkage benchmark, satisfies

$$\frac{1}{J} \sum_{j=1}^J \sigma^2(1 - S_j) = \sigma^2(1 - \bar{S}) = \sigma^2 \left\{ (1 - I) - \frac{\partial H}{\partial m} \right\} = \sigma^2 \{ (1 - I) - (1 - 2I) H \} + O(\rho_3).$$

Proof in Appendix C.

Part (i) is exact: moving the centroid along a fixed profile shape, the rate of change of the dispersion equals the difference between the arithmetic and KL summaries of reliability, and by part (iii) this difference is what separates benchmark risk from the level $\sigma^2(1 - I)$ implied by the centroid alone. Part (ii) is local: to second order, H is a Fisher-information-weighted logit variance, and the difference $\bar{S} - I$ has the sign of $(1 - 2I)$, so for profiles whose centered logits are nearly symmetric and not too spread, modest dispersion at fixed I lowers benchmark risk when $I < 1/2$ and raises it when $I > 1/2$.

The local statement does not hold globally. The $O(\rho_3)$ term depends on the skew of the profile, through a coefficient that itself changes sign along the informativeness scale, and it can dominate: a $J = 25$ profile with one strongly positive logit deviation attains $I = .30$ and $V = 2$ exactly yet has $\bar{S} = .273 < I$, so dispersion of that shape raises benchmark risk on the low- I side of the boundary, and the mirrored shape reverses the sign on the high- I side (Appendix C). The direction of the dispersion effect is therefore a property of the profile, determined exactly by the sign of $\partial H / \partial m = \bar{S} - I$, with dispersion lowering benchmark risk precisely when it lifts $\bar{S}$ above I . It is not a rule keyed to I alone, and the simulation in Section 7 accordingly reports results by profile shape.

A further qualification concerns resources. Holding I fixed while spreading the profile is a statement about summaries: at fixed $\text{GM}(v_j)$, spreading raises total precision $\sum_j 1/v_j$, and Section 7.5 shows that under a fixed enrollment budget, where Proposition 4 applies, imbalance is not beneficial. Rank and distribution goals also respond to dispersion through a channel that no benchmark calculation captures: heterogeneous weights shrink different sites by different amounts, reordering estimates and reshaping their ensemble distribution. For the MSEL goal, part (iii) has an implication for reporting. Because benchmark risk is exactly $\sigma^2(1 - \bar{S})$, and $\bar{S}$ is itself the right-sided centroid of the profile (Appendix B), we report $\bar{S}$ alongside I throughout.

5.3. The Effective Number of Informative Sites. Dispersion measures spread; it does not say whether the reliability mass is carried broadly or by a few dominant sites. Borrowing the effective-sample-size construction of Kish (1965), define

$$J_{\text{eff}} = \frac{(\sum_j S_j)^2}{\sum_j S_j^2} \in [1, J], \quad (9)$$

the number of equally reliable sites that would carry the same reliability mass as the observed profile. $J_{\text{eff}} = J$ exactly when the profile is flat, and J_{eff} approaches 1 as a single site comes to dominate. A low value of J_{eff} indicates a trial whose apparent information rests on a handful of sites, so that conclusions about the effect distribution are sensitive to those particular sites; we treat it as a companion to (I, H) .

5.4. Estimating the Diagnostics. In practice σ^2 is unknown, so S_j , I , and H are functionals of an estimated quantity. There are two natural computations: plug the posterior-mean point estimate of σ^2 into the formulas, or average the functionals over the posterior of σ^2 . The two disagree in a predictable direction.

Proposition 3 (The plug-in estimator overstates informativeness). For fixed sampling variances, $I(\sigma^2) = \sigma^2/(\sigma^2 + \tilde{v})$ is strictly concave in σ^2 . Consequently, for any posterior distribution over σ^2 ,

$$I(\mathbb{E}[\sigma^2 \mid \text{data}]) \geq \mathbb{E}[I(\sigma^2) \mid \text{data}],$$

with strict inequality whenever the posterior is nondegenerate: the plug-in estimate exceeds the posterior-functional estimate. This posterior Jensen gap is approximately $\frac{1}{2}|I''|\text{Var}(\sigma^2 \mid \text{data})$. Under a regular asymptotic sequence, independent sites with the true σ^2 interior and $J^{-1}\sum_j(\sigma^2 + v_j)^{-2}$ bounded away from zero, posterior intervals for I have width $O(J^{-1/2})$ and the gap is $O(J^{-1})$. Proof in Appendix D.

The same concavity argument parallels the small- J optimism documented for heterogeneity statistics such as I^2 (von Hippel, 2015), and it favors a posterior-functional computation, which is also how uncertainty should be attached to the diagnostics: draw $\sigma^{2(m)}$ from its marginal posterior, compute $S_j^{(m)}$, $I^{(m)}$, $H^{(m)}$, and $J_{\text{eff}}^{(m)}$ for each draw, and report posterior means with credible intervals. In the normal-normal model the marginal posterior of σ is one-dimensional and can be evaluated on a grid (Gelman et al., 2013; Rubin, 1981), so the entire diagnostic suite runs in a fraction of a second and requires no Markov chain Monte Carlo. All analyses in Sections 7 and 8 use this computation with a uniform prior on (τ, σ) . Figure 3 collects the workflow, from the site-level inputs to the reported diagnostics, with the empirical analysis of Section 8 carried along as a worked example.

6. Design Implications

The centroid characterization has a planning-stage consequence. Suppose, as in expression (2), that sampling variances scale as $v_j = \kappa/n_j$ for a design constant κ common across sites, and that a total sample of size N is to be allocated.

Proposition 4 (Balanced allocation maximizes informativeness). Under $v_j = \kappa/n_j$ and $\sum_j n_j = N$, the informativeness I is maximized if and only if $n_1 = \dots = n_J = N/J$, in which case $I = \sigma^2/(\sigma^2 + \kappa J/N)$; any unequal allocation is strictly worse, and under integer constraints the maximizer is as equal as possible. Proof and constrained variants in Appendix E.

The proof is a one-line application of the arithmetic-geometric mean inequality, but the result has practical uses. One is that it translates the geometric-mean structure of I into planning guidance: for a fixed total sample size, even recruitment across sites

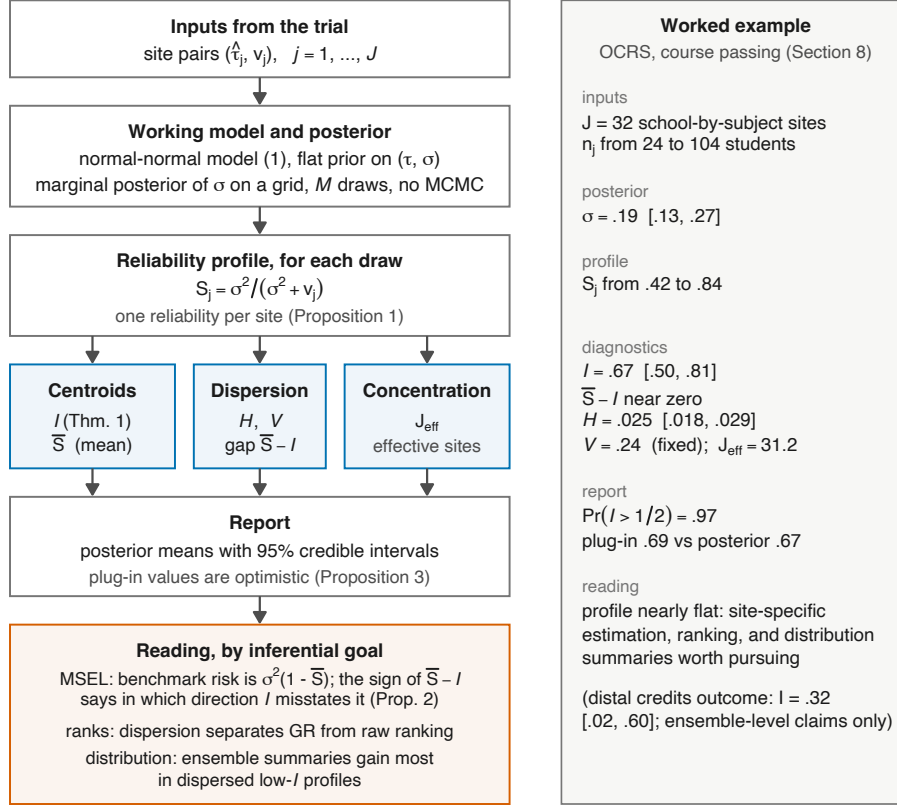


FIGURE 3. Computation and reporting workflow for the reliability-geometry diagnostics. The left column states each step; Appendix D gives the algorithm. The right column carries the workflow through the passing outcome of the Online Credit Recovery Study ($J = 32$; Section 8), whose reported values appear in Table 5.

increases informativeness, and concentrating recruitment in a few large sites reduces it, before any site is analyzed. Another is that it quantifies the cost of imbalance. The effective per-site size entering I is the geometric mean $\tilde{n} = \{\prod_j n_j\}^{1/J}$, and a second-order expansion gives $\tilde{n}/\bar{n} \approx \exp(-CV^2/2)$ for site sizes with coefficient of variation CV ; the approximation is distribution-dependent beyond second order (for log-normal sizes the exact ratio is $(1 + CV^2)^{-1/2}$), and for truncated-Gamma site sizes it is accurate to within about two points over the design range of Lee et al. (2025) (Appendix E). The factor, plotted in Figure 4, explains a finding of Lee et al. (2025): across their 1,500 conditions, varying the coefficient of variation of site sizes from 0 to 0.75 barely moved any performance metric, because $\exp(-0.75^2/2) \approx 0.75$ changes $\tilde{v}$,

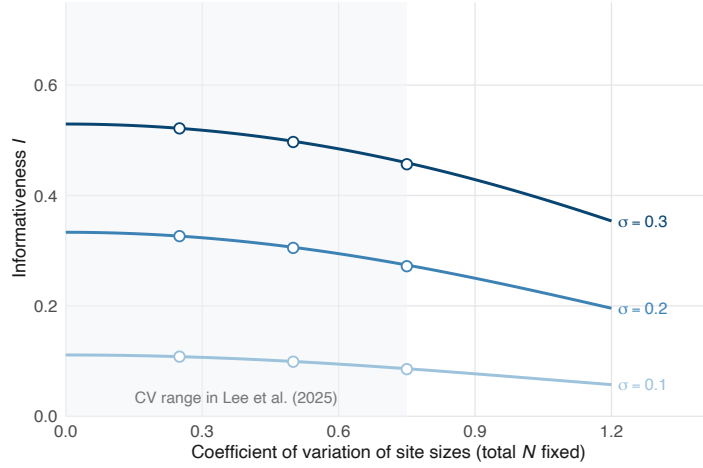


FIGURE 4. Informativeness as a function of site-size imbalance at fixed total sample size ($\kappa = 4$, $\bar{n} = 50$). Curves use the second-order efficiency factor $\tilde{n}/\bar{n} \approx \exp(-CV^2/2)$ that accompanies Proposition 4; open circles are exact geometric means for Gamma-distributed site sizes truncated at $n_j \geq 5$. The shaded band marks the design range studied by Lee et al. (2025), over which imbalance moves I by only a few points, consistent with their finding that it is a second-order design factor.

and hence I , far less than the factor-of-sixteen swings in average site size they crossed it with. Thus imbalance matters only to the extent that it moves the geometric mean.

7. Simulation Study

7.1. Simulation Design. The simulation separates two questions that correspond to two different normalizations. Holding I fixed while spreading the profile asks whether I is a sufficient summary of performance; this is not a design comparison, because at fixed $GM(v_j)$ a more dispersed profile carries more total precision $\sum_j 1/v_j$ (by factors of 1.6 to 3.8 at $V = 2$ across our shapes). Holding total enrollment fixed asks the design question, and there Proposition 4 fixes the direction of the answer. Table 3 therefore crosses two arms with the same latent-effect distributions, losses, and estimators: a stress-test arm whose sampling-variance profiles attain exact (I, V) targets in four shapes, including both skew orientations of the concentrated profile, and a fixed-enrollment arm with Gamma-distributed site sizes at coefficients of variation spanning the education-trial range.

Latent effects are generated with the multisiteDGP package (Lee, 2026); each of the 702 + 108 cells is replicated 500 times with common random numbers pairing every profile within a family, so dispersion contrasts are paired and their Monte

Carlo standard errors are reported alongside them. Posterior means and variances are Rao-Blackwellized, averaging conditional moments over the hyperparameter draws rather than sampled site effects; this matters below, because with equal v_j the exact posterior means are a common positive affine transformation of the raw estimates, and the implementation reproduces that tie identically rather than up to simulation noise.

Two benchmarks accompany the feasible estimators: the known- (τ, σ^2) Gaussian linear-shrinkage rule, whose expected MSEL is exactly $\sigma^2(1 - \bar{S})$ by Proposition 2 and which therefore serves as an implementation check rather than as independent evidence, and a semi-oracle that knows σ^2 but estimates τ , which separates the two hyperparameter-learning channels. Under the non-Gaussian effect distributions the former is the best *linear* rule, not the Bayes rule, and we label it accordingly.

7.2. Effects of Dispersion at Fixed Informativeness. Figure 5 summarizes the stress-test arm for the posterior mean’s RMSE. Each tile is the paired percentage change in loss when the logit-scale dispersion rises from 0 to the indicated level at fixed (I, J, G) , shown separately for the four profile shapes. Dispersion at fixed I moves performance well beyond Monte Carlo noise: cell-mean losses carry median relative Monte Carlo errors of 0.6% (RMSE), 1.1% (MSELP), and 2.5% (ISEL), while the tile magnitudes reach 16% for RMSE and more for the other losses (the triple-goal estimator’s ISEL falls by 43% at $I = .05$, $V = 2$, with a paired standard error of 3 points). For the near-symmetric shapes the sign structure follows the local rule of Proposition 2, with benefits below $I = 1/2$, costs above, and an empirical crossing between $I = .45$ and $I = .60$ for the even and bimodal profiles. The skewed shapes depart from the local rule in opposite directions, as the $O(\rho_3)$ term implies: at $I = .45$ and $V = 2$ the concentrated-precise profile already raises RMSE (+3.1%, standard error 0.2), while the mirrored concentrated-noisy profile still lowers it (−7.0%, standard error 0.2), and in every cell the direction agrees with the sign of $\bar{S} - I$ computed directly from the profile. The direction of the dispersion effect is thus a property of the profile that $\bar{S} - I$ summarizes; I and V alone do not determine it.

The benchmark decomposition identifies the channels. For the known-hyperparameter linear rule the paired RMSE ratios at $V = 2$ run from 0.98 at $I = .05$ to 1.11 at $I = .75$, which is the Proposition-2 wedge alone; the feasible posterior mean’s ratios run from 0.86 to 1.12, and the semi-oracle sits between them (0.92 at $I = .05$). At very low informativeness the additional feasible gain therefore splits roughly evenly between learning τ and learning σ^2 , both of which benefit from

TABLE 3. Simulation design: a stress-test arm with exact (I, V) profiles and a fixed-enrollment arm with realistic site sizes.

Factor	Levels
<i>Stress-test arm (702 cells): exact (I, V), geometric mean of v_j held fixed</i>	
Number of sites J	25, 50, 100
Informativeness I	.05, .15, .30, .45, .60, .75 (exact by construction)
Logit-scale dispersion V	0, 0.5, 1, 2 (exact by construction)
Profile shape ($V > 0$)	even (log-uniform); bimodal; concentrated precise ($\lceil J/5 \rceil$ high-reliability sites); concentrated noisy (the mirrored skew)
<i>Fixed-enrollment arm (108 cells): $N = J\bar{n}$ fixed, $v_j = 4/n_j$, Gamma site sizes</i>	
Number of sites J ; mean size $\bar{n}$	$J \in \{25, 50, 100\}$; $\bar{n} \in \{20, 50, 100\}$
Site-size imbalance CV	0, .25, .50, .75 (truncated at $n_j \geq 5$, rescaled to total N)
<i>Common to both arms</i>	
True effect distribution G	Gaussian; two-component Gaussian mixture; asymmetric Laplace (skewed), matching Lee et al. (2025)
Cross-site SD; replications	$\sigma = 0.20$ (effect-size units; diagnostics are scale-free); 500 per cell, with common random numbers pairing all profiles within each (J, I, G) or $(J, \bar{n}, G)$ family
Estimators	ML (no pooling); PM, CB, GR from the fully Bayesian normal-normal model with Rao-Blackwellized posterior moments; two benchmarks for channel decomposition: known- (τ, σ^2) Gaussian linear shrinkage and known- σ^2 semi-oracle
Losses	RMSE, MSEL, ISEL (finite-population)

Note. Stress-test profiles set $\ln v_j = \ln \tilde{v} + x_j$ with $J^{-1} \sum_j x_j = 0$ and $J^{-1} \sum_j x_j^2 = V$, so each cell attains its target (I, V) exactly; holding $\tilde{v}$ fixed while spreading the x_j raises total precision $\sum_j 1/v_j$ (by factors up to 3.8 at $V = 2$), so this arm tests whether I is a sufficient performance summary, not what a designer gains by dispersing a fixed budget; the fixed-enrollment arm answers the design question. Latent effects are generated with the **multisiteDGP** package (Lee, 2026); models are fit by full Bayes with a flat prior on (τ, σ) using the one-dimensional grid method (Appendix D) with $M = 2,000$ posterior draws. ML = maximum likelihood (raw estimates); PM = posterior mean; CB = constrained Bayes; GR = triple-goal.

the precise sites a dispersed profile contains. For rank and distribution losses the profile acts through a further channel that no risk identity captures, the site-varying reordering of estimates, and those effects are largest where the ensemble-corrected summaries operate (Section 7.4).

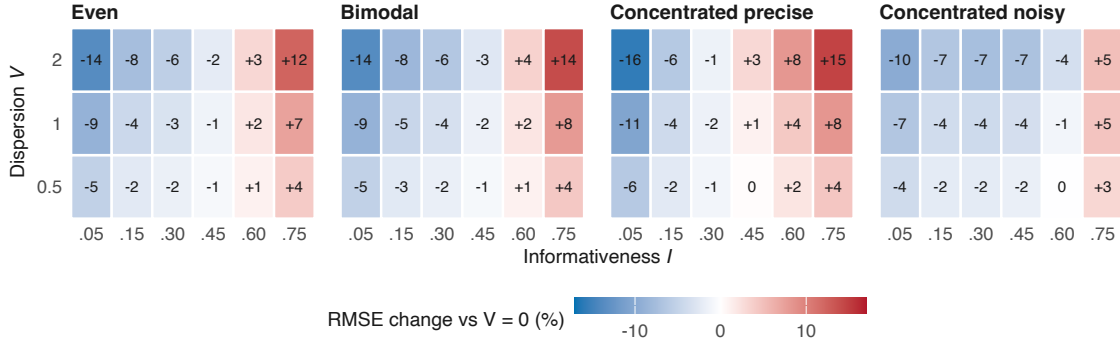


FIGURE 5. Paired percentage change in the posterior mean’s RMSE when dispersion increases at fixed informativeness, by profile shape. The value printed in each tile compares a dispersed profile with the homogeneous profile ($V = 0$) that shares the same informativeness, number of sites, and effect distribution, averaged over J and G ; blue indicates lower loss than under the homogeneous profile and red higher loss. The near-symmetric shapes (left two panels) cross near the $I = 1/2$ boundary of Proposition 2; the two skewed shapes (right two panels) shift the crossing in opposite directions, in accordance with the sign of $\bar{S} - I$ for each profile. Median Monte Carlo error per tile is well under one point.

7.3. Comparison of Single Profile Summaries. Table 4 addresses two versions of the question of what the profile adds beyond I . Panel A bounds what any profile summary could add: augmenting a fully saturated model in (I, J, G) with the complete profile design, the crossed dispersion-by-shape factors, recovers 15 to 52% of the residual log-loss variance for the shrinkage-based estimators (up to 99.9% for the unpooled estimator, whose risk is mechanically the arithmetic mean of the v_j); much of the remainder is the Monte Carlo floor. Panel B compares the candidate single summaries on equal terms: each of the dispersion H , the logit variance V , and the arithmetic mean $\bar{S}$ enters the same model with the same degrees of freedom and the same interaction structure, and is scored out of sample. The arithmetic mean performs best in eight of the nine goal-by-estimator comparisons, by clear margins for MSEL and ranking, which is what Proposition 2 part (iii) leads one to expect for risk-type losses; the exception is the triple-goal estimator’s distribution loss, where H is best by a small margin, consistent with its sensitivity to the higher-order profile structure that ensemble recovery exploits. We do not conclude that H is the single number to attach to I . Rather, the profile’s left and right centroids should be

TABLE 4. Residual explanatory value of the reliability profile beyond informativeness, and a cross-validated comparison of single profile summaries.

<i>Panel A: residual share of cell-mean log-loss variance explained by the full profile design</i>					
Loss		PM	CB	GR	ML
RMSE		.16	.15	.15	>.99
MSELP		.32	.32	.32	.59
ISEL		.22	.36	.52	.76
<i>Panel B: five-fold cross-validated R^2, one summary at a time with identical structure</i>					
Loss	Estimator	+ H	+ V	+ $\bar{S}$	Best
RMSE	PM	.98964	.99155	.99843	$\bar{S}$
	CB	.99099	.99336	.99816	$\bar{S}$
	GR	.99076	.99308	.99784	$\bar{S}$
MSELP	PM	.99269	.99452	.99923	$\bar{S}$
	CB	.99269	.99452	.99923	$\bar{S}$
	GR	.99298	.99470	.99919	$\bar{S}$
ISEL	PM	.99598	.99663	.99757	$\bar{S}$
	CB	.99389	.99575	.99693	$\bar{S}$
	GR	.99586	.99572	.99518	H

Note. Panel A: each entry is $(R_{\text{full}}^2 - R_{\text{base}}^2)/(1 - R_{\text{base}}^2)$, where the base model is fully saturated in $\text{factor}(I) \times \text{factor}(J) \times \text{factor}(G)$ (base R^2 between .943 and .994) and the full model adds the crossed profile factors $\text{factor}(V) \times \text{factor}(\text{shape})$; the unexplained remainder includes the Monte Carlo floor (median relative Monte Carlo error of cell means: 0.6–2.5%). Panel B: each candidate summary $X \in \{H, V, \bar{S}\}$ enters the same base model as $X + X \times \text{factor}(I)$, so all three columns spend identical degrees of freedom; entries are five-fold cross-validated R^2 with folds drawn over cells. $\bar{S}$ performs best in eight of nine comparisons, as Proposition 2 predicts for risk-type losses; H is best, narrowly, for the triple-goal estimator’s distribution loss. PM = posterior mean; CB = constrained Bayes; GR = triple-goal; ML = unpooled estimates.

reported together, because their gap $\bar{S} - I = \partial H / \partial m$ is simultaneously the best-predicting scalar correction and the exact risk difference of Proposition 2; H or V describes how dispersed the profile is, and the shape of the profile accounts for the remainder.

7.4. Dispersion and the Choice of Estimator. For RMSE the posterior mean is best in every cell, as theory requires. For ranking, the baseline comparison is analytic: with equal sampling variances the posterior means and constrained Bayes estimates are a common increasing affine transformation of the raw estimates, so their ranks tie with the raw ranks identically, and the implementation reproduces

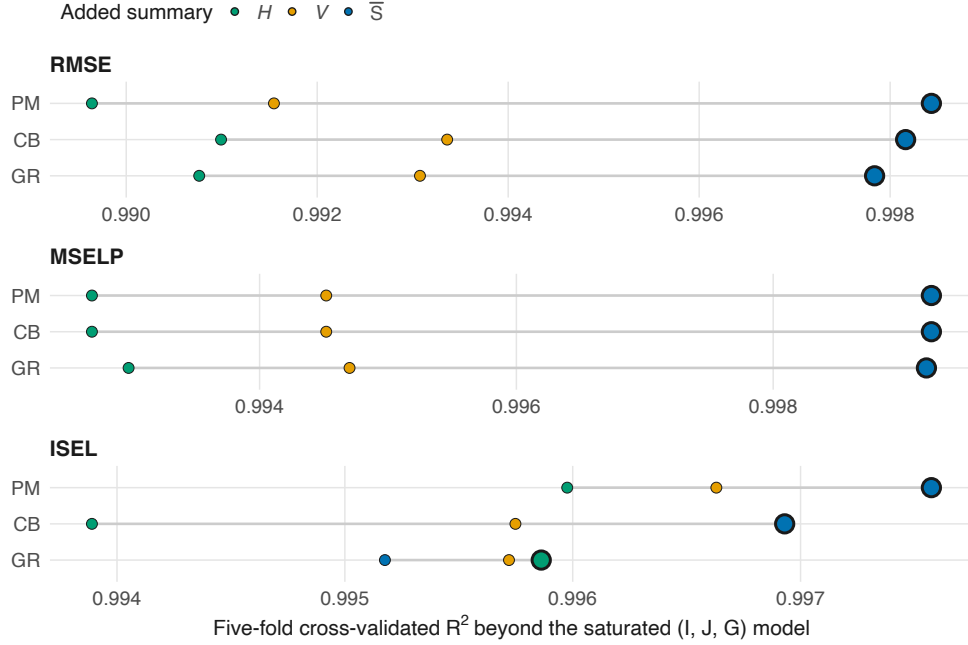


FIGURE 6. Five-fold cross-validated R^2 for adding each candidate profile summary, with identical degrees of freedom and interaction structure, to a saturated (I, J, G) meta-model of cell-mean log losses. Each row is one goal-by-estimator comparison; the filled marker identifies the summary with the highest cross-validated R^2 in that row. The arithmetic mean $\bar{S}$ is best in eight of the nine comparisons, as Proposition 2 leads one to expect for risk-type losses; the dispersion H is best, by a small margin, for the triple-goal estimator’s distribution loss.

the tie exactly. Winner counts in homogeneous cells, which an earlier version of this simulation produced, are Monte Carlo noise around that tie. Dispersion breaks the equivalence, and Figure 7 shows the consequence: the triple-goal estimator’s expected-rank construction separates from the raw ranking as V grows, with paired MSEL gains over the unpooled estimator of essentially zero at $V = 0$ rising to 8 to 17% at $V = 2$, largest in the middle of the informativeness range. For the distribution goal, the ensemble-corrected summaries dominate the posterior mean throughout, and their advantage widens in dispersed low- I profiles, where a few nearly unshrunk sites allow the spread of the ensemble to be recovered. The reliability profile thus indicates, in advance, whether model-based ranking and ensemble correction are likely to be worthwhile, which I alone cannot indicate.

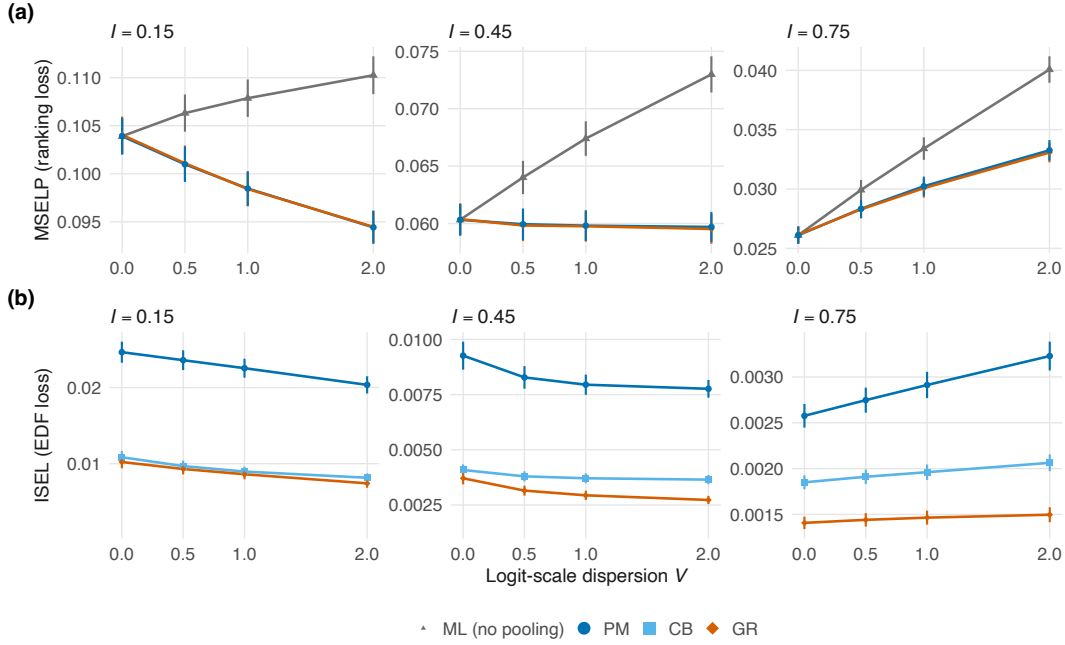


FIGURE 7. Ranking and distribution losses as dispersion increases at fixed informativeness ($J = 50$, Gaussian effects, even profiles; error bars are 95% Monte Carlo intervals for cell means). Panel (a): ranking loss (MSEL) for the unpooled (ML), posterior-mean (PM), and triple-goal (GR) estimators; at $V = 0$ the PM and CB ranks tie with the ML ranks exactly, and GR separates as the profile disperses. Panel (b): distribution loss (ISEL) for PM, CB, and GR; the ensemble-corrected summaries widen their advantage in dispersed low- I profiles.

7.5. The Fixed-Enrollment Arm. The fixed-enrollment arm returns to Proposition 4. With $N = J\bar{n}$ held fixed and Gamma site sizes at $CV \in \{0, .25, .50, .75\}$, imbalance simultaneously lowers I (from .333 to .279 on average at $\sigma = 0.20$, $\kappa = 4$) and raises V (to .61), and every loss deteriorates monotonically: at $CV = .75$ versus balanced allocation, the posterior mean's RMSE is 3% higher, its ranking loss 8% higher, and its distribution loss 11% higher, with the same ordering for every estimator. We found no condition under which imbalance improved any loss at a fixed budget. The two arms thus answer different questions: the stress-test arm shows that I is an incomplete summary of a given profile, and the fixed-enrollment arm shows that a designer who can still choose the profile should balance, in accordance with Proposition 4. Robustness across the three effect distributions, per-shape ratio tables with Monte Carlo standard errors, and the fixed-enrollment tables are given in Appendix F.

8. Empirical Illustration: The Online Credit Recovery Study

8.1. Data and Site-Level Estimates. The Online Credit Recovery Study randomized first-year high school students in a large urban district who had failed Algebra 1 or English 9 to either an online summer credit-recovery course or the school’s usual teacher-directed course, within blocks defined by school, subject, and cohort, in the summers of 2018 and 2019 (American Institutes for Research, 2024; Rickles et al., 2024); it is the successor to an earlier Chicago trial of the same contrast (Heppen et al., 2017). The public-use impact file contains 1,737 student-level records. Crossing the 24 schools with the two subjects yields $J = 32$ sites in which both arms are present, with site sizes from 24 to 104 students (median 49).

Because six sites span two cohorts with treated shares that differ across cohorts by up to .31, each site’s intent-to-treat effect on the study’s primary proximal outcome, passing the credit-recovery course, is estimated by within-site least squares on the assignment indicator with randomization-block fixed effects and an HC2 standard error, rather than by a pooled difference in means; the adjustment moves individual site estimates by up to .24 (median .03). A covariate-adjusted version and a subject-matched distal outcome, credits earned by the end of year four, serve as sensitivity and contrast analyses; all outcomes used are observed for every student. The site estimates vary widely around a pooled average of $-.130$: as in the published study, the online format reduced course passing on average, with substantial cross-site heterogeneity ($\sigma = .19$, 95% CI $[-.13, .27]$).

8.2. Diagnostic Results and Sensitivity Analyses. Table 5 and Figure 8 report the diagnostics, computed as posterior functionals. For the proximal outcome the trial is highly informative for an education trial: $I = .67$ $[-.50, .81]$ with $\Pr(I > 1/2 \mid \text{data}) = .97$, because cross-site variation in true effects is large relative to within-site noise. The dispersion is low ($H = .025$, $V = .24$, $\bar{S} - I$ near zero, $J_{\text{eff}} = 31.2$ of 32), so the centroid is an adequate one-number summary of the profile, and site-specific estimation, ranking, and distributional summaries may be worthwhile. The distal credits outcome lies in the opposite regime: informativeness drops to $I = .32$ with a 95% interval spanning $[-.02, .60]$ and $\Pr(I > 1/2) = .11$, and with 32 sites the diagnostics themselves are estimated with considerable uncertainty. In every analysis the plug-in estimate exceeds the posterior-functional estimate, .69 against .67 for passing and .35 against .32 for credits, in the direction Proposition 3 predicts.

Because Section 7 shows that conclusions of this kind can be fragile, we accompany the diagnostics with three sensitivity analyses (Appendix G). First, eight schools contribute both an Algebra and an English site, so the pooled likelihood overstates independence. Refitting all 2^8 one-site-per-paired-school selections moves the passing I across $[\text{.61}, \text{.75}]$ and the credits I across $[\text{.11}, \text{.43}]$; the contrast between the two regimes survives, though the pooled interval endpoints should be read cautiously, and the subject-specific fits in Table 5 avoid the duplication at the cost of answering subject-specific questions. Second, on the log-odds scale the passing analysis gives $I = .53 [\text{.25}, \text{.75}]$ with $\Pr(I > 1/2) = .62$. The position of a binary outcome relative to any fixed reference value thus depends on the effect scale, and we attach the conclusions to the risk-difference estimand chosen a priori rather than to the trial as such. Third, the sampling variances are estimates based on 8 to 53 students per arm. A student bootstrap within arms and blocks widens the intervals accordingly, to roughly $[\text{.66}, \text{.83}]$ for the passing I , $[\text{.38}, \text{.73}]$ for the credits I , and $[\text{.19}, \text{.32}]$ for the passing V , whose treatment as conditionally fixed understates its variability. The conclusion we draw is modest: course passing is the stronger candidate for site-specific analysis, and the distal outcome supports little beyond the pooled average; this distinction survived every sensitivity analysis we conducted. The diagnostics do not by themselves validate individual site ranks or distributional claims, for which rank-uncertainty methods are available (Gu & Koenker, 2023; Mogstad et al., 2024).

9. Discussion

This paper set out to answer two questions about the informativeness index that Lee et al. (2025) showed governs what multisite trials can say about their sites: why the geometric mean appears in the index, and what the average conceals. The two questions have a common answer. Site shrinkage weights are reliability coefficients in the projection sense of Zumbo (2026a), so a multisite trial carries a profile of measurement-quality coefficients on $(0, 1)$; the informativeness index is the centroid of that profile in the Kullback–Leibler geometry natural to that scale, which is where the geometric mean comes from (Theorem 1); and because centroids are projections, the index comes with a residual, the dispersion H , that measures what any one-number summary discards (Theorem 2). The derivative of H along the informativeness scale equals the gap between the reported index and the arithmetic reliability summary that governs benchmark risk (Proposition 2); the sign of that gap is $(1 - 2I)$ for near-symmetric profiles and otherwise depends on the profile’s

TABLE 5. Reliability-geometry diagnostics for the Online Credit Recovery Study.

Analysis	J	$\hat{\tau}$	σ [95% CI]	I [95% CI]	Plug-in $\hat{I}$	$\Pr(I > \frac{1}{2})$	H [95% CI]	J_{eff}
<i>Proximal outcome: passed the credit-recovery course (risk difference; block-adjusted)</i>								
Pooled	32	-.130	.19 [.13, .27]	.67 [.50, .81]	.69	.97	.025 [.018, .029]	31.2
Algebra 1	13	-.080	.19 [.09, .34]	.63 [.29, .87]	.69	.81	.016 [.009, .020]	12.7
English 9	19	-.163	.20 [.12, .32]	.71 [.49, .87]	.74	.97	.028 [.018, .034]	18.5
Adjusted	32	-.117	.18 [.12, .26]	.65 [.48, .80]	.67	.96	.028 [.020, .032]	31.0
<i>Distal outcome: subject-matched credits earned by end of year 4 (block-adjusted)</i>								
Pooled	32	-.121	2.80 [0.60, 4.95]	.32 [.02, .60]	.35	.11	.028 [.003, .038]	28.8

Note. Sites are school-by-subject cells; every site enrolled both arms (n_j from 24 to 104, median 49). Site effects are within-site intent-to-treat regressions on the assignment indicator with randomization-block fixed effects and HC2 standard errors; the adjusted row additionally controls for the grade-8 state test score and ninth-grade GPA (imputed, with missingness indicators). All diagnostics are posterior functionals under the normal-normal model with a flat prior on (τ, σ) , using 20,000 draws; plug-in $\hat{I}$ evaluates $I(\cdot)$ at the posterior mean of σ^2 and exceeds the posterior-functional estimate in every row (Proposition 3). The logit-scale profile variance V is .24 (pooled passing), .16 (Algebra), .28 (English), .26 (adjusted), and .31 (credits). Credits are measured in district credit units (observed range 0 to 90 in these data), so σ for the distal outcome is not comparable in magnitude to the proximal rows; the scale-free diagnostics are comparable. Dependence, effect-scale, and first-stage sensitivity analyses appear in Appendix G. CI = credible interval.

skew. In the simulations, the profile predicted estimator performance where informativeness could not, and it indicated in advance when model-based ranking separates from raw ranking and when ensemble-corrected distribution estimates improve on the posterior means. A balanced-allocation result (Proposition 4) and the plug-in bias (Proposition 3) complete the set of results, and the Online Credit Recovery Study illustrates the full workflow, which runs in seconds on public data.

The diagnostics are intended to complement existing practice. The I^2 statistic of Higgins and Thompson (2002) measures whether heterogeneity is present; the pooling factors of Gelman and Pardoe (2006) describe how strongly each unit is pulled toward the mean; I , H , and J_{eff} describe how recoverable the site-specific structure is, on average and in profile. For reporting, we recommend the following: alongside the caterpillar plot that most multisite analyses already show, report the reliability profile with its two centroids, the informativeness I and the mean reliability $\bar{S}$, together with the dispersion (H or the logit-scale variance V), with credible intervals from the posterior-functional computation; J_{eff} is an optional descriptive companion. When the dispersion is small, the analyst may treat the trial as having a single reliability and

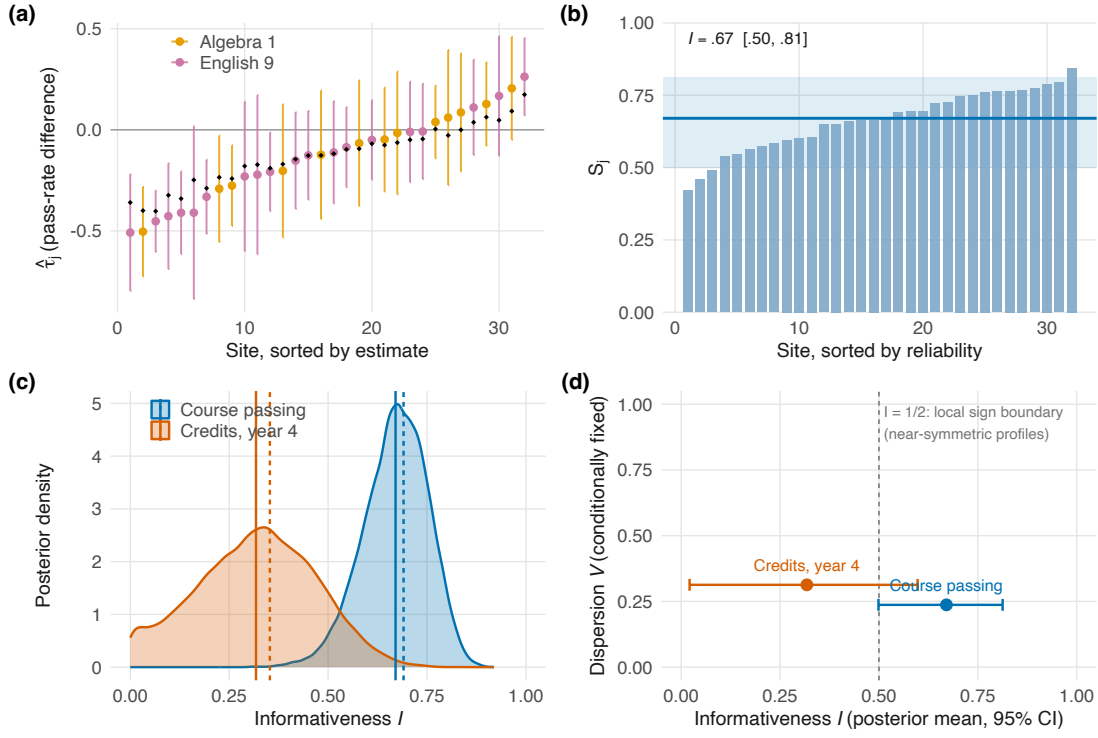


FIGURE 8. Reliability-geometry diagnostics for the Online Credit Recovery Study ($J = 32$ school-by-subject sites; block-adjusted site effects). (a) Within-site intent-to-treat effects on course passing with 95% intervals, colored by subject; black diamonds are posterior means. (b) Posterior-mean site reliabilities, sorted, with the posterior mean and 95% band for I . (c) Posterior distributions of I for the proximal (passing) and distal (credits) outcomes; solid lines mark posterior-functional means and dashed lines the plug-in estimates, which are optimistic in both cases (Proposition 3). (d) The two analyses on the (I, V) plane; the dashed line marks the local sign boundary of Proposition 2, which applies to near-symmetric profiles at the risk-difference effect scale used here.

rely on the guidance of Lee et al. (2025) directly. When it is large, the profile rather than the centroid should be the object of interest: the sign of $\bar{S} - I$ indicates the direction in which I misstates benchmark estimation risk, and the estimator comparisons of Section 7 indicate which summaries warrant sensitivity analysis. The diagnostics indicate which analyses are worth pursuing; they do not by themselves justify conclusions about individual ranks, for which purpose-built inference is available (Gu & Koenker, 2023; Mogstad et al., 2024).

The results are subject to limitations. One is that the sampling variances are treated as known throughout, as they are in the index we set out to explain; with very small sites the uncertainty in v_j propagates into the profile, and a chi-square correction would be needed. Another is the assumption that the true effects and their sampling variances are unrelated. When precision predicts effects, which has been documented in some education settings, the marginal reliability profile misstates the information geometry; the conditional location-scale approach of Chen (2026) indicates how the diagnostics could be generalized, and we regard a precision-dependent informativeness as the natural next step. A third is that our simulations fit the Gaussian working model even when the true effect distribution was not Gaussian, matching how the diagnostics would be computed in practice, but we did not evaluate semiparametric priors; the interaction between profile dispersion and flexible-prior performance is an open question. Finally, boundary behavior requires care: a single site with near-zero sampling variance drives I toward one even though the remaining sites are unchanged, while the logit-scale dispersion V diverges and H approaches a positive limit that records how far the other sites sit from perfect reliability; the joint report (I, H, J_{eff}) is intended so that no single number is relied on alone in such profiles.

Extensions of the geometry are developed elsewhere. The centroid-and-residual pair used here for diagnosis suggests using the same Bregman machinery for estimation, replacing the Euclidean projection implicit in constrained Bayes with a divergence matched to the inferential goal. In addition, the computation of Section 5.4, together with reference implementations of the diagnostics, is being packaged for routine use. More generally, the quantities that multisite analysts already compute arise from a simple geometry. Making the geometry explicit yields a characterization of the informativeness index, a dispersion diagnostic with a risk interpretation, and a design rule, and it places trials on a common scale on which they can be compared before any model is fitted.

Software and reproducibility. The diagnostics, the grid-based posterior-functional computation of Appendix D, the two simulation arms, and the empirical analysis were computed in R. Latent effects for the simulation are generated with the `multisiteDGP` package (Lee, 2026) (<https://github.com/joonho112/multisiteDGP>). A replication package with the ordered analysis scripts, the verification suite V1–V13, and the scripts that regenerate every figure and table is available at <https://github.com/joonho112/geometry-of-informativeness-replication>; every

number reported in Sections 7 and 8 and in Appendices F and G regenerates from it. The analysis pipeline aborts if any verification block or structural invariant fails.

Data availability. The empirical illustration uses the public-use impact-analysis file of the Online Credit Recovery Study (American Institutes for Research, 2024), available from openICPSR project 202181 (<https://www.openicpsr.org/openicpsr/project/202181>). The data belong to the original study team and to that release and are not redistributed with the replication code; the replication scripts read the data through the `OCRS_DATA` environment variable.

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Online Supplement

The Geometry of Informativeness: A Bregman-Centroid Diagnostic for Partial Pooling in Multisite Trials

JoonHo Lee

Contents

Appendix A. Projection Reliability and the Proof of Proposition 1	33
A.1. Conditional Expectation as Orthogonal Projection	33
A.2. Proof of Proposition 1	33
A.3. Boundary Cases and Assumptions	34
A.4. Remark on the Measurement Reading	35
Appendix B. Bregman Centroids and the Proof of Theorem 1	35
B.1. Bregman Divergences and Sided Centroids	35
B.2. The Bernoulli–KL Geometry on $(0, 1)$	36
B.3. Proof of Theorem 1	36
B.4. Remarks	37
Appendix C. Dispersion: Proofs of Theorem 2 and Proposition 2	38
C.1. Proof of Theorem 2	38
C.2. Proof of Proposition 2	39
C.3. Discussion of the Expansion	40
Appendix D. Estimation of the Diagnostics: Proof of Proposition 3 and Computation	41
D.1. Proof of Proposition 3	41
D.2. Posterior-Functional Computation	42
Appendix E. Design: Proof of Proposition 4 and Constrained Variants	43
E.1. Proof of Proposition 4	43
E.2. The Efficiency Factor $\exp(-CV^2/2)$	43
E.3. Constrained and Cost-Weighted Variants	44
Appendix F. Simulation Details and Additional Results	45
F.1. Constructing Profiles with Exact (I, V)	45
F.2. Fitting, Estimators, and Numerical Assurance	46
F.3. Stress-Test Results in Full	47

F.4. The Fixed-Enrollment Arm	47
F.5. Reproducibility	47
Appendix G. Details of the Online Credit Recovery Study Analysis	49
G.1. Data and Site Definition	49
G.2. Outcomes	50
G.3. Block-Adjusted Site-Level Estimation	50
G.4. Model Fitting and Diagnostics	50
G.5. Sensitivity Analyses	51
G.6. Replication	52

Appendix A. Projection Reliability and the Proof of Proposition 1

This appendix records the Hilbert-space facts behind the reliability reading of the shrinkage weight and proves Proposition 1 of the main text, including its boundary cases. The material is standard; we collect it so that the main text’s claims are self-contained. Throughout, $L^2 = L^2(\Omega, \mathcal{F}, P)$ denotes the space of square-integrable random variables with inner product $\langle X, Y \rangle = \mathbb{E}(XY)$.

A.1. Conditional Expectation as Orthogonal Projection. Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra. The conditional expectation $\mathbb{E}[\cdot \mid \mathcal{G}]$ is the orthogonal projection of $L^2(\Omega, \mathcal{F}, P)$ onto the closed subspace $L^2(\Omega, \mathcal{G}, P)$: for $X \in L^2$, the true score $T = \mathbb{E}[X \mid \mathcal{G}]$ is the unique $\mathcal{G}$ -measurable minimizer of $\|X - Z\|^2$ over $Z \in L^2(\mathcal{G})$, and the residual $E = X - T$ is orthogonal to every element of $L^2(\mathcal{G})$ (e.g., Zumbo, 2026a, Appendix B). Orthogonality delivers the variance decomposition

$$\text{Var}(X) = \text{Var}(T) + \mathbb{E}[(X - T)^2], \quad (\text{A.1})$$

and the reliability of X with respect to $\mathcal{G}$ is defined as the projection efficiency

$$\text{Rel}(X) = \frac{\text{Var}(T)}{\text{Var}(X)} = \cos^2(\theta_{\tilde{X}, \tilde{T}}), \quad (\text{A.2})$$

where $\tilde{X} = X - \mathbb{E}X$, $\tilde{T} = T - \mathbb{E}T$, and θ is the angle between them in L^2 (Zumbo, 2026a). Definition (A.2) makes no reference to parallel forms or hypothetical repeated administration: it requires an observed variable, a conditioning σ -algebra naming the object of measurement, and a joint law.

A.2. Proof of Proposition 1.

Proof. Work under model (1) and set $X_j = \hat{\tau}_j$ and $\mathcal{G}_j = \sigma(\tau_j)$. The first stage of the model states $\hat{\tau}_j \mid \tau_j \sim \mathbf{N}(\tau_j, v_j)$, so the projection is

$$T_j = \mathbb{E}[\hat{\tau}_j \mid \tau_j] = \tau_j,$$

which uses only the conditional mean, not normality. The second stage gives $\text{Var}(T_j) = \text{Var}(\tau_j) = \sigma^2$. By the law of total variance, or directly from (A.1) with $\mathbb{E}[(X_j - T_j)^2] = \mathbb{E}\{\text{Var}(\hat{\tau}_j \mid \tau_j)\} = v_j$,

$$\text{Var}(\hat{\tau}_j) = \sigma^2 + v_j.$$

Substituting into (A.2),

$$\text{Rel}(\hat{\tau}_j) = \frac{\sigma^2}{\sigma^2 + v_j} = S_j,$$

which is the shrinkage weight in equation (4). For the angle identity, orthogonality of T_j and $E_j = X_j - T_j$ gives $\langle \tilde{X}_j, \tilde{T}_j \rangle = \text{Var}(T_j)$, hence

$$\cos^2 \theta_j = \frac{\langle \tilde{X}_j, \tilde{T}_j \rangle^2}{\|\tilde{X}_j\|^2 \|\tilde{T}_j\|^2} = \frac{\text{Var}(T_j)^2}{\text{Var}(X_j) \text{Var}(T_j)} = \frac{\sigma^2}{\sigma^2 + v_j} = S_j,$$

and consequently $\omega_j = 1 - S_j = \sin^2 \theta_j$. The pooling-factor identification follows from the definition of Gelman and Pardoe (2006): for the random-coefficient component τ_j of model (1), their pooling factor is the ratio of the sampling variance to the marginal variance of $\hat{\tau}_j$, which is $v_j/(\sigma^2 + v_j) = \omega_j$. $\square$

A.3. Boundary Cases and Assumptions. Three degenerate regimes are recorded here because the diagnostics of the main text inherit their behavior.

- (a) *Perfectly measured site* ($v_j \rightarrow 0$). Then $S_j \rightarrow 1$, $\theta_j \rightarrow 0$: the observed estimate coincides with the true effect. If a single site, say site 1, attains $v_1 \rightarrow 0$ while $\sigma^2 > 0$, the geometric mean $\tilde{v}$ tends to 0 and $I \rightarrow 1$ even though the remaining sites may be arbitrarily noisy. This boundary sensitivity of geometric-mean aggregation is flagged by the logit-scale dispersion V , which diverges in the same limit; the left-sided dispersion H remains bounded, converging to $J^{-1} \sum_{j \neq 1} \ln(1/S_j)$, a positive record of how far the unchanged sites sit from perfect reliability. Both behaviors are reasons to report (I, H, J_{eff}) , with V , jointly rather than I alone.
- (b) *No cross-site variation* ($\sigma^2 \rightarrow 0$). Then every $S_j \rightarrow 0$ and $I \rightarrow 0$: complete pooling is optimal and the trial carries no information about site-specific departures from the common effect. The diagnostics are conventionally set to their limits, with $H \rightarrow 0$.

(c) *Assumptions.* Proposition 1 requires exactly three conditions: first-stage conditional unbiasedness, $\mathbb{E}[\hat{\tau}_j \mid \tau_j] = \tau_j$; the first-stage variance condition $\mathbb{E}\{\text{Var}(\hat{\tau}_j \mid \tau_j)\} = v_j$, which gives the error term its advertised size; and $\text{Var}(\tau_j) = \sigma^2 \in (0, \infty)$. The result is site-by-site, so no cross-site independence is needed, and normality in either stage is not used. What is used, here and throughout the paper, is that v_j is known and functionally unrelated to τ_j ; violations of the latter, called precision dependence, are discussed in Section 9.

A.4. Remark on the Measurement Reading. The identification of sites as objects of measurement places the multisite trial inside generalizability theory: the design is a one-facet G-study in which the within-site experiment plays the role of the measurement occasion, universe-score variance is σ^2 , and error variance is v_j (Brennan, 2001; Cronbach et al., 1972). Under this reading, S_j is a generalizability coefficient computed at site-specific error variance. The reading matters because the questions of what is being measured, and by what instrument, have precise answers here: the measurand is the site-average treatment effect τ_j , the instrument is the randomized experiment run at site j , and the replication is supplied by the randomization distribution itself. Classical test theory defines reliability through hypothetical repeated administrations (Lord & Novick, 1968); here none need to be imagined, and the projection definition of Zumbo (2026a) applies directly.

Appendix B. Bregman Centroids and the Proof of Theorem 1

B.1. Bregman Divergences and Sided Centroids. Let ϕ be a strictly convex, differentiable potential on an open interval $\mathcal{X} \subseteq \mathbb{R}$. The Bregman divergence generated by ϕ is

$$D_\phi(x, y) = \phi(x) - \phi(y) - \phi'(y)(x - y), \quad x, y \in \mathcal{X}, \quad (\text{B.1})$$

the excess of $\phi(x)$ over its tangent-line approximation at y ; strict convexity gives $D_\phi(x, y) \geq 0$ with equality if and only if $x = y$ (Banerjee et al., 2005; Bregman, 1967). The divergence is asymmetric in general, so a cloud of points $S_1, \dots, S_J \in \mathcal{X}$ has two natural centers (Nielsen & Nock, 2009). The *right centroid* minimizes divergence in the second argument and is always the arithmetic mean,

$$\arg \min_{c \in \mathcal{X}} \frac{1}{J} \sum_{j=1}^J D_\phi(S_j, c) = \bar{S} = \frac{1}{J} \sum_{j=1}^J S_j, \quad (\text{B.2})$$

a result of Banerjee et al. (2005, Proposition 1). The *left centroid* minimizes divergence in the first argument,

$$I_\phi = \arg \min_{c \in \mathcal{X}} \frac{1}{J} \sum_{j=1}^J D_\phi(c, S_j), \quad (\text{B.3})$$

and averages in the dual coordinates ϕ' : differentiating under the sum, $\partial_c D_\phi(c, S_j) = \phi'(c) - \phi'(S_j)$, so the unique stationary point satisfies

$$\phi'(I_\phi) = \frac{1}{J} \sum_{j=1}^J \phi'(S_j) \quad \Longleftrightarrow \quad I_\phi = (\phi')^{-1} \left\{ \frac{1}{J} \sum_{j=1}^J \phi'(S_j) \right\}, \quad (\text{B.4})$$

which is a minimum because the objective has second derivative $\phi''(c) > 0$. Equation (B.4) is the dual-coordinate linearity that Zumbo (2026b) places at the center of the information-geometric extension of test theory: Bregman projections are linear in dual coordinates and nonlinear in primal ones. The projection and duality machinery is classical convex analysis (Banerjee et al., 2005; Bregman, 1967; Nielsen & Nock, 2009), and its test-theoretic reading is due to Zumbo (2026b); as far as we are aware, the application of sided centroids to aggregate a profile of reliability coefficients, which is the subject of Theorem 1, is new here.

B.2. The Bernoulli–KL Geometry on $(0, 1)$. Take $\mathcal{X} = (0, 1)$ and the negative Bernoulli entropy

$$\begin{aligned} \phi_{\text{KL}}(p) &= p \ln p + (1 - p) \ln(1 - p), \\ \phi'_{\text{KL}}(p) &= \ln \frac{p}{1 - p} = \text{logit}(p), \quad \phi''_{\text{KL}}(p) = \frac{1}{p(1 - p)}. \end{aligned} \quad (\text{B.5})$$

Substituting into (B.1) gives, for $p, q \in (0, 1)$,

$$D_{\phi_{\text{KL}}}(p, q) = p \ln \frac{p}{q} + (1 - p) \ln \frac{1 - p}{1 - q} = \text{KL}\{\text{Bern}(p) \parallel \text{Bern}(q)\}, \quad (\text{B.6})$$

the Kullback–Leibler divergence between Bernoulli laws. The dual coordinate is the logit, which is also the natural parameter of the Bernoulli exponential family, and ϕ'' is the reciprocal Fisher information; these identifications are used in Appendix C.

B.3. Proof of Theorem 1.

Proof. Existence and uniqueness. The objective $F(p) = J^{-1} \sum_j \text{KL}\{\text{Bern}(p) \parallel \text{Bern}(S_j)\}$ has derivative

$$F'(p) = \text{logit}(p) - \frac{1}{J} \sum_{j=1}^J \text{logit}(S_j)$$

and second derivative $F''(p) = \{p(1-p)\}^{-1} > 0$, so F is strictly convex on $(0, 1)$; moreover $F'(p) \rightarrow -\infty$ as $p \downarrow 0$ and $F'(p) \rightarrow +\infty$ as $p \uparrow 1$, so the unique minimizer is interior and solves $F'(p) = 0$:

$$\text{logit}(I) = \frac{1}{J} \sum_{j=1}^J \text{logit}(S_j), \quad (\text{B.7})$$

which is (B.4) for the potential (B.5).

Identification with the index of Lee et al. (2025). Because $S_j = \sigma^2/(\sigma^2 + v_j)$,

$$\text{logit}(S_j) = \ln \frac{S_j}{1-S_j} = \ln \frac{\sigma^2}{v_j} = \ln \sigma^2 - \ln v_j,$$

so the right-hand side of (B.7) equals $\ln \sigma^2 - J^{-1} \sum_j \ln v_j$. Applying the inverse logit,

$$I = \frac{1}{1 + \exp\{-\ln \sigma^2 + J^{-1} \sum_j \ln v_j\}} = \frac{\sigma^2}{\sigma^2 + \exp\{J^{-1} \sum_j \ln v_j\}} = \frac{\sigma^2}{\sigma^2 + \tilde{v}},$$

with $\tilde{v}$ the geometric mean of $v_1, \dots, v_J$. This is equation (5), the informativeness of Lee et al. (2025). $\square$

B.4. Remarks.

- (a) *Why the left centroid.* By (B.2), the right KL centroid of the reliabilities is the arithmetic mean $\bar{S}$, so the sided distinction is exactly the distinction between the two candidate “average reliabilities” of the main text, and the index of Lee et al. (2025) is the left one. The left centroid is the value p whose Bernoulli law best approximates every site’s law simultaneously in reverse KL; operationally, it averages on the natural-parameter (logit) scale, which is why it responds to ratios of signal to noise, σ^2/v_j , rather than to their levels.
- (b) *Scale equivariance.* Both sides of (B.7) are invariant to multiplying all of $(\sigma^2, v_1, \dots, v_J)$ by a common constant, so I is a pure signal-to-noise summary; the same is true of H , V , and J_{eff} .
- (c) *Degenerate profiles.* If a single S_j approaches 0 or 1 while every other reliability remains strictly interior and fixed, the average logit diverges and I moves to that boundary; Appendix A(a) discusses the practical reading. Opposing endpoint limits need not reach a boundary: if $S_1 = \epsilon$ and $S_2 = 1 - \epsilon$, the two logits cancel and $I = 1/2$ for every ϵ , and if one reliability is exactly 0 and another exactly 1 the average reverse-KL objective is infinite at every interior p , so a joint limiting value depends on the path. The verification code includes the cancellation example.

Appendix C. Dispersion: Proofs of Theorem 2 and Proposition 2

C.1. Proof of Theorem 2.

Proof. Let ϕ be strictly convex and differentiable on $\mathcal{X}$, let I_ϕ denote the left centroid (B.3), and fix any $p \in \mathcal{X}$. Expanding definitions,

$$\frac{1}{J} \sum_{j=1}^J D_\phi(p, S_j) = \phi(p) - \frac{1}{J} \sum_j \phi(S_j) - \frac{1}{J} \sum_j \phi'(S_j)(p - S_j), \quad (\text{C.1})$$

$$\begin{aligned} D_\phi(p, I_\phi) + \frac{1}{J} \sum_{j=1}^J D_\phi(I_\phi, S_j) &= \phi(p) - \phi'(I_\phi)(p - I_\phi) - \frac{1}{J} \sum_j \phi(S_j) \\ &\quad - \frac{1}{J} \sum_j \phi'(S_j)(I_\phi - S_j), \end{aligned} \quad (\text{C.2})$$

where the $\phi(I_\phi)$ terms in (C.2) cancel. Subtracting (C.2) from (C.1) leaves

$$\frac{1}{J} \sum_{j=1}^J D_\phi(p, S_j) - \left\{ D_\phi(p, I_\phi) + H_\phi \right\} = (p - I_\phi) \left\{ \phi'(I_\phi) - \frac{1}{J} \sum_j \phi'(S_j) \right\}, \quad (\text{C.3})$$

and the brace vanishes by the dual-mean condition (B.4), proving the decomposition for every p .

For the converse, suppose the display holds for all p with some center $c \in \mathcal{X}$ and some constant h in place of (I_ϕ, H_ϕ) . Evaluating the resulting identity at two points $p \neq p'$ and differencing gives

$$-\frac{1}{J} \sum_j \phi'(S_j) (p - p') = -\phi'(c) (p - p'),$$

so $\phi'(c) = J^{-1} \sum_j \phi'(S_j)$, and strict monotonicity of ϕ' forces $c = I_\phi$; setting $p = c$ then forces $h = H_\phi$. Under $\phi = \phi_{\text{KL}}$ the statement of the main text follows with $H = H_{\phi_{\text{KL}}}$. $\square$

Equation (C.3) shows what is special about the left centroid: it is the unique center for which the cross term vanishes for every reference point, the Bregman analogue of the fact that sums of squared deviations decompose exactly around the arithmetic mean. The decomposition is a constant-manifold instance of the generalized Pythagorean identity for Bregman projections (Bregman, 1967; Nielsen & Nock, 2009; Zumbo, 2026b): I_ϕ is the Bregman projection of the profile onto the diagonal $\{(p, \dots, p)\}$, and H_ϕ is its squared residual in divergence units.

C.2. Proof of Proposition 2. Write $L_j = \text{logit}(S_j)$, $\bar{L} = J^{-1} \sum_j L_j = \text{logit}(I)$ by Theorem 1, $\Delta_j = L_j - \bar{L}$, $V = J^{-1} \sum_j \Delta_j^2$, and $\rho_3 = J^{-1} \sum_j |\Delta_j|^3$. Let $\sigma(\cdot)$ denote the logistic function, so $S_j = \sigma(L_j)$ and $I = \sigma(\bar{L})$.

Proof. Part (i), the exact identity. With $I = \sigma(m)$ and $S_j = \sigma(m + \Delta_j)$, use $\ln \sigma(x) = x - \ln(1 + e^x)$ and $\ln\{1 - \sigma(x)\} = -\ln(1 + e^x)$ to expand each divergence:

$$\begin{aligned} \text{KL}\{\text{Bern}(I) \parallel \text{Bern}(S_j)\} &= I \{\ln I - \ln S_j\} + (1 - I) \{\ln(1 - I) - \ln(1 - S_j)\} \\ &= -I\Delta_j + \ln(1 + e^{m+\Delta_j}) - \ln(1 + e^m). \end{aligned}$$

Averaging over j and using $J^{-1} \sum_j \Delta_j = 0$ gives the closed form

$$H(m; \Delta) = \frac{1}{J} \sum_{j=1}^J \ln(1 + e^{m+\Delta_j}) - \ln(1 + e^m), \quad (\text{C.4})$$

and differentiating in m , with the centered profile Δ held fixed,

$$\frac{\partial H}{\partial m} = \frac{1}{J} \sum_{j=1}^J \sigma(m + \Delta_j) - \sigma(m) = \bar{S} - I,$$

which is the exact identity of part (i).

Part (ii), the local expansion. Define $f(L) = \text{KL}\{\text{Bern}(I) \parallel \text{Bern}(\sigma(L))\}$. Using $\ln \sigma(L) = L - \ln(1 + e^L)$ and $\ln\{1 - \sigma(L)\} = -\ln(1 + e^L)$,

$$f(L) = \phi_{\text{KL}}(I) - IL + \ln(1 + e^L),$$

so that

$$\begin{aligned} f'(L) &= \sigma(L) - I, & f''(L) &= \sigma(L)\{1 - \sigma(L)\}, \\ f'''(L) &= \sigma(L)\{1 - \sigma(L)\}\{1 - 2\sigma(L)\}. \end{aligned}$$

At $L = \bar{L}$: $f(\bar{L}) = 0$, $f'(\bar{L}) = 0$, and $f''(\bar{L}) = I(1 - I)$; the vanishing of f' at the centroid is the first-order condition of Theorem 1 read in logit coordinates. A third-order Taylor expansion with Lagrange remainder gives, for each j ,

$$f(L_j) = \frac{1}{2}I(1 - I)\Delta_j^2 + \frac{1}{6}f'''(\xi_j)\Delta_j^3$$

for some ξ_j between $\bar{L}$ and L_j . Averaging over j and using the global bound $\sup_L |f'''(L)| = \sup_{u \in (0,1)} |u(1-u)(1-2u)| = \sqrt{3}/18$,

$$|H - \frac{1}{2}I(1 - I)V| \leq \frac{\sqrt{3}}{108} \rho_3. \quad (\text{C.5})$$

Because $f''(\bar{L}) = I(1 - I)$ is the Fisher information of the Bernoulli family at its natural parameter $\bar{L}$, the leading term is the Fisher-weighted logit variance claimed in the main text.

For the companion expansion of the gap, expand $\bar{S} = J^{-1} \sum_j \sigma(L_j)$ around $\bar{L}$, with $\sigma' = \sigma(1 - \sigma)$, $\sigma'' = \sigma(1 - \sigma)(1 - 2\sigma)$, and $\sup_L |\sigma'''(L)| = 1/8$,

$$|\bar{S} - I - \tfrac{1}{2}I(1 - I)(1 - 2I)V| \leq \frac{1}{48}\rho_3, \quad (\text{C.6})$$

where the linear term vanishes because $J^{-1} \sum_j \Delta_j = 0$. Combining (C.5) and (C.6),

$$\bar{S} - I = (1 - 2I)H + R, \quad |R| \leq \left(\frac{1}{48} + \frac{\sqrt{3}}{108} \right) \rho_3 < \frac{\rho_3}{25},$$

which completes part (ii) with an explicit remainder constant.

Part (iii). With known hyperparameters (τ, σ^2) , the posterior of τ_j is $\mathbf{N}\{\tau_j^*, \sigma^2(1 - S_j)\}$ by equation (3), so the integrated risk of the oracle posterior mean at site j is $\mathbb{E}\{(\tau_j^* - \tau_j)^2\} = \sigma^2(1 - S_j)$ by the law of total expectation, and averaging over sites gives the exact identity

$$\frac{1}{J} \sum_{j=1}^J \sigma^2(1 - S_j) = \sigma^2(1 - \bar{S}).$$

Substituting part (ii) yields $\sigma^2(1 - \bar{S}) = \sigma^2\{(1 - I) - (1 - 2I)H\} + O(\rho_3)$, completing the proof. $\square$

C.3. Discussion of the Expansion. Three comments on the scope of Proposition 2.

- (a) *The design constant V .* Because $L_j = \ln \sigma^2 - \ln v_j$, the logit-scale variance V equals the variance of $\{\ln v_j\}$ and does not involve σ^2 at all. When the sampling variances are treated as known, V is a design constant carrying no posterior uncertainty, while $H \approx \frac{1}{2}I(1 - I)V$ inherits uncertainty only through the factor $I(1 - I)$. This division is visible in the empirical illustration of the main text, where the credible interval for V is degenerate and that for H is narrow relative to the interval for I .
- (b) *The sign of the dispersion effect is a property of the profile.* By part (i), the direction of the dispersion effect on benchmark risk at fixed I is set by $\bar{S} - I = \partial H / \partial m$, with dispersion lowering the risk exactly when it lifts $\bar{S}$ above I ; part (ii) shows the sign of $\bar{S} - I$ is that of $(1 - 2I)$ only up to the $O(\rho_3)$ skew term. The skew term can dominate inside practically relevant ranges. Take $J = 25$, $m = \text{logit}(.30)$, and centered deviations $\Delta_1 = \sqrt{48}$,

$\Delta_2 = \dots = \Delta_{25} = -\sqrt{1/12}$: then $J^{-1} \sum_j \Delta_j = 0$ and $J^{-1} \sum_j \Delta_j^2 = 2$ exactly, so $I = .30$ and $V = 2$, yet $\bar{S} = .2732 < I$, and dispersion of this shape raises benchmark risk on the low- I side; the mirrored profile reverses the conclusion on the high- I side. The concentrated profile of Appendix F, with a minority of high-reliability sites, behaves the same way in sign at $I = .45$. All statements in the main text are therefore local (near-symmetric profiles, I bounded away from $1/2$) or shape-conditional, and the simulation reports results by profile shape. This example is included in the verification code.

- (c) *Accuracy.* The remainder bounds involve third moments of the logit profile and are conservative. Numerically, the relative error of approximation (i) is below 3% for even profiles with $V \leq 0.4$, and the ratio $H/\{\frac{1}{2}I(1-I)V\}$ converges to 1 as $V \rightarrow 0$; verification code accompanies the reproducibility materials (Appendix F).
- (d) *Scope of the risk link.* Part (iii) concerns the oracle posterior-mean estimator and the MSEL goal. It isolates one channel through which dispersion moves performance at fixed I : the wedge between the arithmetic and KL summaries of the profile. With estimated hyperparameters a second channel opens, because dispersed profiles contain high-reliability sites that sharpen the estimation of σ^2 ; and for rank and distribution goals a third channel opens, because site-varying weights reorder and reshape the ensemble of estimates. The simulation in the main text measures the total effect of all channels; Appendix F separates the oracle channel by reporting the oracle posterior mean alongside the feasible estimators.

Appendix D. Estimation of the Diagnostics: Proof of Proposition 3 and Computation

D.1. Proof of Proposition 3.

Proof. Fix the sampling variances and write $c = \tilde{v} > 0$ for their geometric mean, so $I(\sigma^2) = \sigma^2/(\sigma^2 + c)$ as a function of the heterogeneity variance. Differentiating twice,

$$I'(\sigma^2) = \frac{c}{(\sigma^2 + c)^2} > 0, \quad I''(\sigma^2) = \frac{-2c}{(\sigma^2 + c)^3} < 0,$$

so $I(\cdot)$ is strictly increasing and strictly concave on $(0, \infty)$. Jensen's inequality applied to the posterior distribution of σ^2 gives

$$\mathbb{E}[I(\sigma^2) \mid \text{data}] \leq I(\mathbb{E}[\sigma^2 \mid \text{data}]),$$

with equality only for a degenerate posterior, which proves the overstatement claim. The same argument applies site by site to $S_j(\sigma^2) = \sigma^2/(\sigma^2 + v_j)$, so plug-in site reliabilities are individually overstated as well.

For the rate claim, the Fisher information for σ^2 in the marginal model $\hat{\tau}_j \sim \mathbf{N}(\tau, \sigma^2 + v_j)$ is $\mathcal{I}(\sigma^2) = \frac{1}{2} \sum_j (\sigma^2 + v_j)^{-2}$, whether τ is known or estimated, because the mean and variance blocks of the normal information matrix are orthogonal. For profiles with v_j uniformly bounded and the true σ^2 interior, $\mathcal{I}$ grows linearly in J ; hence, by the usual Bernstein–von Mises argument, the posterior standard deviation of σ^2 , and by the delta method that of $I(\sigma^2)$, is $O(J^{-1/2})$, and the Jensen gap $I(\mathbb{E}[\sigma^2 | \cdot]) - \mathbb{E}[I(\sigma^2) | \cdot] \approx \frac{1}{2} |I''| \text{Var}(\sigma^2 | \cdot)$ is $O(J^{-1})$. $\square$

The direction of the bias parallels the small-sample optimism documented for the heterogeneity statistic I^2 by von Hippel (2015): both phenomena are Jensen gaps of concave variance-ratio functionals evaluated at noisy variance estimates, and both matter most when the number of sites or studies is small; this is also the setting in which a single plug-in number is most often reported.

D.2. Posterior-Functional Computation. All diagnostics reported in the main text are computed as posterior functionals under the normal-normal model with a flat prior on τ and a flat prior on σ truncated to $[0, \sigma_{\max}]$; $\sigma_{\max}$ is chosen adaptively (and enlarged until the top of the grid carries mass below 10^{-4}), so the computation is numerical truncation of the improper flat prior rather than a fixed proper uniform, and draws are jittered uniformly within grid cells. Against fine deterministic quadrature these choices move posterior functionals of I by at most about 10^{-3} in low-informativeness test cases. The computation requires no Markov chain Monte Carlo (Gelman et al., 2013; Rubin, 1981):

1. Evaluate the marginal log posterior of σ on a fine grid: with $w_j(\sigma) = (\sigma^2 + v_j)^{-1}$ and $\hat{\mu}(\sigma) = \sum_j w_j \hat{\tau}_j / \sum_j w_j$,
$$\ln p(\sigma | \text{data}) = \text{const} - \frac{1}{2} \ln \sum_j w_j(\sigma) + \sum_j \left\{ -\frac{1}{2} \ln(\sigma^2 + v_j) - \frac{1}{2} w_j(\sigma) \{ \hat{\tau}_j - \hat{\mu}(\sigma) \}^2 \right\}.$$
2. Sample $\sigma^{(1)}, \dots, \sigma^{(M)}$ from the normalized grid (with within-cell jitter), then $\tau^{(m)} | \sigma^{(m)} \sim \mathbf{N}\{\hat{\mu}(\sigma^{(m)}), (\sum_j w_j)^{-1}\}$ and, when site effects are needed, $\tau_j^{(m)}$ from equation (3).
3. For each draw, form $S_j^{(m)} = \sigma^{2(m)} / (\sigma^{2(m)} + v_j)$ and evaluate the diagnostics
$$I^{(m)} = \text{logistic} \left\{ \frac{1}{J} \sum_j \text{logit } S_j^{(m)} \right\}, \quad H^{(m)} = \frac{1}{J} \sum_j \text{KL}\{\text{Bern}(I^{(m)}) \| \text{Bern}(S_j^{(m)})\},$$

$$J_{\text{eff}}^{(m)} = \frac{(\sum_j S_j^{(m)})^2}{\sum_j S_j^{(m)2}}.$$

4. Report posterior means and central credible intervals of $\{I^{(m)}\}$, $\{H^{(m)}\}$, $\{J_{\text{eff}}^{(m)}\}$, and, if desired, of each $S_j^{(m)}$. The logit-scale profile variance V requires no propagation because it is free of σ^2 (Appendix C.3, remark (a)).

With a grid of 250 points and $M = 20,000$ draws the full suite runs in well under a second for J in the hundreds. When analysts prefer a different fitting engine, for example Markov chain Monte Carlo under half-Cauchy priors or restricted maximum likelihood with a parametric bootstrap, step 3 applies unchanged to whatever draws of σ^2 the engine produces; the diagnostics are functionals of the heterogeneity posterior rather than of the fitting engine.

Appendix E. Design: Proof of Proposition 4 and Constrained Variants

E.1. Proof of Proposition 4.

Proof. Under $v_j = \kappa/n_j$ the geometric mean of the sampling variances is

$$\tilde{v} = \exp\left\{\frac{1}{J} \sum_j \ln \frac{\kappa}{n_j}\right\} = \frac{\kappa}{\tilde{n}}, \quad \tilde{n} = \left(\prod_j n_j\right)^{1/J},$$

so $I = \sigma^2/(\sigma^2 + \kappa/\tilde{n})$ is strictly increasing in the geometric mean site size $\tilde{n}$. By the arithmetic-geometric mean inequality, subject to $\sum_j n_j = N$,

$$\tilde{n} \leq \frac{1}{J} \sum_j n_j = \frac{N}{J},$$

with equality if and only if $n_1 = \dots = n_J$. Hence I is uniquely maximized by the balanced allocation, with maximum $I = \sigma^2/(\sigma^2 + \kappa J/N)$, and any unequal allocation is strictly worse. Equivalently, maximizing I is the convex program $\max \sum_j \ln n_j$ subject to $\sum_j n_j = N$, whose stationarity condition $1/n_j = \lambda$ forces equality.

For the integer-constrained problem, note that if two sites differ by at least 2, say $n_a \geq n_b + 2$, the transfer $n_a \mapsto n_a - 1$, $n_b \mapsto n_b + 1$ strictly increases $n_a n_b$, hence $\tilde{n}$; iterating, any maximizer satisfies $\max_j n_j - \min_j n_j \leq 1$, that is, $n_j \in \{\lfloor N/J \rfloor, \lceil N/J \rceil\}$. $\square$

E.2. The Efficiency Factor $\exp(-CV^2/2)$. For site sizes with mean $\bar{n}$ and coefficient of variation CV , a second-order expansion of $\ln n$ around $\bar{n}$ gives $\mathbb{E} \ln n \approx$

$\ln \bar{n} - \text{CV}^2/2$, hence

$$\frac{\tilde{n}}{\bar{n}} \approx \exp\left(-\frac{\text{CV}^2}{2}\right),$$

to second order in general; for log-normal sizes the exact ratio is $(1 + \text{CV}^2)^{-1/2}$, which the display matches to the same order (the display is exact for log-normal sizes when CV^2 is read as the log-scale variance). The informativeness under imbalance is therefore that of a balanced trial whose per-site size is deflated by this factor: $\text{CV} = 0.25, 0.50, 0.75$ deflate the effective site size by about 3%, 12%, and 25%. Because Lee et al. (2025) crossed these CV values with average site sizes spanning a factor of sixteen, the geometry predicts, and their meta-model regressions found, that CV is a second-order design factor: imbalance matters only through the modest movement it induces in the geometric mean. For the site-size distributions they used (Gamma with the stated CV, truncated at $n_j \geq 5$), the approximation above is accurate to within about half a point through $\text{CV} = 0.5$ and about two points at $\text{CV} = 0.75$; verification code accompanies Appendix F.

E.3. Constrained and Cost-Weighted Variants. Two extensions used in planning practice preserve the structure of the proof.

- (a) *Box constraints.* If sites carry bounds $n_j \in [L_j, U_j]$ with $\sum_j L_j \leq N \leq \sum_j U_j$, the maximizer of $\sum_j \ln n_j$ is the water-filling solution: $n_j = \min\{U_j, \max(L_j, 1/\lambda)\}$ with λ chosen so the budget binds; when no bound is active it reduces to N/J .
- (b) *Cost-weighted budgets.* If enrolling one unit at site j costs w_j and the constraint is $\sum_j w_j n_j = B$, the Lagrange condition $1/n_j = \lambda w_j$ gives $n_j^* = B/(Jw_j)$: allocate inversely to unit cost, spending an equal share of budget at every site. Equal allocation is recovered when costs are constant.
- (c) *Dispersion at the design stage.* Under the scaling $v_j = \kappa/n_j$ with common κ , the balanced design also sets the logit-scale profile variance V , and hence the anticipated dispersion H , to zero: balance simultaneously maximizes the centroid and eliminates the dispersion. When κ varies across sites, for example because outcome variances or treatment shares differ, the two objectives can separate, and a penalized criterion such as maximizing $I - \lambda H$ over feasible allocations provides a simple way to trade the two objectives against each other; we do not pursue optimality theory for it here.

Appendix F. Simulation Details and Additional Results

F.1. Constructing Profiles with Exact (I, V) . Every stress-test cell fixes the informativeness and the logit-scale dispersion exactly, rather than in expectation. Given targets (I, V) , a cross-site standard deviation $\sigma = 0.20$, and J sites, set the target geometric mean $\tilde{v} = \sigma^2(1 - I)/I$ and construct

$$\ln v_j = \ln \tilde{v} + x_j, \quad \frac{1}{J} \sum_j x_j = 0, \quad \frac{1}{J} \sum_j x_j^2 = V, \quad (\text{F.1})$$

so that $\text{logit}(S_j) = \ln \sigma^2 - \ln v_j$ has mean $\text{logit}(I)$ and variance V by construction. Four shapes for $\{x_j\}$ are used:

- (a) *Even*: x_j proportional to an equally spaced grid, standardized to mean zero and mean square V (a log-uniform ladder of sampling variances);
- (b) *Bimodal*: half the sites at $-\sqrt{V}$, half at $+\sqrt{V}$;
- (c) *Concentrated precise*: $k = \lceil J/5 \rceil$ precise sites at $x = -\sqrt{V(J-k)/k}$ and $J-k$ diffuse sites at $x = +\sqrt{Vk/(J-k)}$, so a fifth of the sites carry reliabilities far above the rest;
- (d) *Concentrated noisy*: the mirrored skew, k noisy sites at $x = +\sqrt{V(J-k)/k}$ with the remainder at $x = -\sqrt{Vk/(J-k)}$. Both orientations are included because at fixed (I, V) the difference $\bar{S} - I$, and with it the direction of the dispersion effect on benchmark risk, depends on the skew of the logit profile jointly with I : mirrored skews cross the $\bar{S} = I$ boundary at different points of the informativeness scale (Appendix C).

At $V = 0$ the shapes coincide and are simulated once. Holding $\tilde{v}$ fixed while spreading the x_j raises total precision $\sum_j 1/v_j$ by $J^{-1} \sum_j e^{-x_j} \geq 1$; at $V = 2$ the factor is 2.35 (even), 2.18 (bimodal), 3.78 (concentrated precise), and 1.63 (concentrated noisy) at $J = 50$. The stress-test arm therefore tests whether I suffices as a performance summary; the design comparison at fixed resources is the subject of the fixed-enrollment arm of Section F.4. Latent effects $\tau_j = \sigma z_j$ use the standardized generators of the `multisiteDGP` package (version 0.1.0, numerically identical to the public 0.1.1 release): Gaussian; the two-component Gaussian mixture with separation $\delta = 5$, weight $\varepsilon = .3$, and scale ratio $\Upsilon = 2$; and the asymmetric Laplace with quantile parameter $\rho = .3$, matching the three population shapes of Lee et al. (2025). Observed estimates are $\hat{\tau}_j \sim \mathbf{N}(\tau_j, v_j)$. Common random numbers are used throughout: within each (J, I, G) family, the latent-effect matrix and the standard-normal error matrix are drawn once

and shared across all thirteen (V, shape) combinations, so every dispersion contrast is paired.

F.2. Fitting, Estimators, and Numerical Assurance. Each replication is fit by the grid-based full-Bayes computation of Appendix D with $M = 2,000$ joint posterior draws. Posterior means and variances are *Rao-Blackwellized*: with conditional moments $\mu_j(\tau, \sigma) = \tau + S_j(\sigma)(\hat{\tau}_j - \tau)$ and $s_j^2(\sigma) = \sigma^2\{1 - S_j(\sigma)\}$, the reported moments are $\widehat{\text{pm}}_j = M^{-1} \sum_m \mu_j^{(m)}$ and, by the law of total variance, $\widehat{\text{pv}}_j = M^{-1} \sum_m s_j^{2(m)} + \widehat{\text{Var}}_m(\mu_j^{(m)})$. Because each $\mu_j^{(m)}$ is affine in $\hat{\tau}_j$ with a positive slope, equal sampling variances make $\widehat{\text{pm}}$ a common increasing affine transformation of the raw estimates, so posterior-mean and constrained-Bayes ranks tie with the raw ranks identically in homogeneous profiles; the analysis pipeline asserts this invariant on every run and aborts if it fails (the maximum absolute MSELP discrepancy across all 27,000 homogeneous replications is zero to machine precision). An earlier draw-based implementation violated the tie by Monte Carlo noise and produced spurious homogeneous-cell “winners”; no such quantity is reported here. Constrained Bayes rescales the Rao-Blackwellized means so the finite-population variance matches $\hat{\sigma}^2 = J^{-1} \sum_j \widehat{\text{pv}}_j + \text{Var}_{\text{fp}}(\widehat{\text{pm}})$, the standard large- J simplification of the posterior expected finite-population variance (Ghosh, 1992; Louis, 1984); the triple-goal (GR) estimates place the J mid-quantiles of the pooled posterior draws at the ranks of the posterior expected ranks (Shen & Louis, 1998), a step that remains draw-based and whose Monte Carlo error is included in the reported standard errors. Two benchmarks complete the roster: **opm**, the known- (τ, σ^2) *Gaussian linear-shrinkage benchmark* $\tau + S_j(\hat{\tau}_j - \tau)$, equal to $S_j \hat{\tau}_j$ here because the standardized generators set $\tau = 0$ (under the mixture and asymmetric-Laplace generators this is the best linear rule, not the Bayes rule), and **spm**, the semi-oracle with σ^2 known and τ estimated, $S_j \hat{\tau}_j + (1 - S_j) \hat{\mu}$. Losses follow equations (5)–(8) of Lee et al. (2025): RMSE, MSELP with average-tie ranks, and ISEL computed exactly as the integrated squared difference of the two empirical distribution functions. Verification blocks V1–V13, distributed with the replication code and exiting with nonzero status on any failure, confirm the formal statements numerically, including the exact identity $\partial H / \partial m = \bar{S} - I$, the profile-dependent sign counterexamples of Appendix C, the two-site endpoint cancellation, the homogeneous rank-tie invariant, and a Monte Carlo check of Proposition 1, alongside the Rubin (1981) eight-schools benchmark, the CB calibration, and the ISEL integrator against brute-force numerical integration.

F.3. Stress-Test Results in Full. Table F.1 reports paired $V = 2$ versus $V = 0$ loss ratios for the even profile, all estimators and benchmarks; Table F.2 reports the posterior mean’s RMSE ratios by profile shape; and Table F.3 shows the triple-goal distribution-loss ratios by effect distribution. Monte Carlo standard errors, computed from family-level paired log-ratios, are below .014 for every RMSE ratio in Tables F.1 and F.2, below .029 for MSEL, and below .067 for ISEL (the largest attaching to the unpooled estimator’s high- I entries); cell-mean losses carry median relative Monte Carlo errors of 0.6%, 1.1%, and 2.5% respectively. The tables show several features beyond those discussed in the main text. One is that the unpooled estimator is uniformly harmed by dispersion in RMSE (ratios near 1.5 throughout), because its risk is the arithmetic mean of the v_j , which rises with dispersion at fixed geometric mean; the beneficial dispersion effects at low I are produced by shrinkage. Another is that the constrained-Bayes ranking row is identical to the posterior-mean row by construction (the two are increasing affine transformations of each other), a consequence of the construction rather than a finding. A third is the channel split: the benchmark decomposition divides the feasible posterior mean’s extra low- I gain roughly evenly between the two learning channels, with ratios at $I = .05$ of .976 (both hyperparameters known), .919 (σ^2 known, τ estimated), and .863 (both estimated). Finally, the ensemble-corrected summaries gain the most from dispersion in ISEL at low informativeness (CB .49, GR .57 at $I = .05$), because a dispersed profile contains nearly unshrunk sites from which the spread of the ensemble can be recovered.

F.4. The Fixed-Enrollment Arm. With total enrollment $N = J\bar{n}$ fixed, $v_j = 4/n_j$, and Gamma site sizes truncated at $n_j \geq 5$ and rescaled to sum to N , increasing the coefficient of variation of site sizes lowers I and raises V simultaneously, and every loss deteriorates monotonically for every estimator (Table F.4). This is the design-relevant normalization, and the results agree with Proposition 4: at a fixed budget we found no beneficial dispersion.

F.5. Reproducibility. All simulations, verification blocks, and tables are regenerated by the scripts 00_utils.R through 08_sim_fixedN.R distributed with the replication materials, and the figures by the figure scripts distributed alongside them, with fixed seeds throughout. The two simulation arms reported here were executed on a 32-vCPU cloud node (AWS EC2 c6a.8xlarge, R 4.6.0) in about thirty minutes end to end, including the verification suite and the empirical analysis of Appendix G;

TABLE F.1. Paired loss ratios, $V = 2$ versus $V = 0$ at fixed (I, J, G) : even profiles, averaged over J and G with common random numbers.

Estimator, loss	Informativeness I					
	.05	.15	.30	.45	.60	.75
PM, RMSE	.86	.92	.94	.98	1.03	1.12
CB, RMSE	.76	.88	.96	1.00	1.06	1.15
GR, RMSE	.76	.89	.96	1.01	1.05	1.13
ML, RMSE	1.51	1.51	1.50	1.51	1.50	1.50
Linear benchmark, RMSE	.98	.95	.96	.98	1.03	1.11
Semi-oracle, RMSE	.92	.94	.95	.98	1.03	1.12
PM (= CB), MSEL	.93	.92	.94	.99	1.07	1.20
GR, MSEL	.93	.92	.94	.99	1.06	1.18
ML, MSEL	1.02	1.05	1.11	1.19	1.28	1.40
PM, ISEL	.98	.88	.84	.92	1.05	1.20
CB, ISEL	.49	.76	.83	.89	.96	1.07
GR, ISEL	.57	.77	.76	.82	.88	.98
ML, ISEL	1.07	1.19	1.34	1.45	1.52	1.55

TABLE F.2. The direction of the dispersion effect is a shape property: posterior-mean RMSE ratios ($V = 2$ vs. $V = 0$) by profile shape.

Profile shape	Informativeness I					
	.05	.15	.30	.45	.60	.75
Even	.86	.92	.94	.98	1.03	1.12
Bimodal	.86	.92	.94	.97	1.04	1.14
Concentrated precise	.84	.94	.99	1.03	1.08	1.15
Concentrated noisy	.90	.93	.93	.93	.96	1.05

Note. The near-symmetric shapes cross near $I = 1/2$; the concentrated-precise profile ($\bar{S} < I$) crosses between .30 and .45, and the concentrated-noisy profile ($\bar{S} > I$) not until past .60. In every cell the direction agrees with the sign of $\bar{S} - I$ computed from the profile, the exact statement of Proposition 2(i). Paired Monte Carlo standard errors are at most .014.

`sessionInfo` snapshots for every run are archived alongside the outputs, and the analysis pipeline aborts if any verification block or structural invariant fails.

TABLE F.3. Robustness across effect distributions: triple-goal ISEL ratios ($V = 2$ vs. $V = 0$, even profiles).

True G	Informativeness I					
	.05	.15	.30	.45	.60	.75
Gaussian	.56	.71	.69	.81	.96	1.06
Gaussian mixture	.64	.79	.78	.79	.82	.89
Asymmetric Laplace	.52	.80	.82	.85	.87	.99

Note. The direction and approximate magnitude of the dispersion effect are shared across the three effect distributions; under the bimodal mixture the beneficial region extends past $I = .5$ because distribution-shape recovery continues to benefit from nearly unshrunk sites.

TABLE F.4. Fixed-enrollment arm: paired loss ratios relative to balanced allocation, averaged over J , $\bar{n}$, and G .

CV	Posterior mean			Triple-goal			Realized diagnostics	
	RMSE	MSELP	ISEL	RMSE	MSELP	ISEL	I	V
.25	1.00	1.01	1.01	1.01	1.01	1.01	.327	.06
.50	1.02	1.04	1.05	1.03	1.04	1.03	.308	.27
.75	1.03	1.08	1.11	1.05	1.08	1.05	.279	.61

Note. Balanced allocation ($CV = 0$) has $I = .333$ and $V = 0$ at these settings ($\sigma = 0.20$, $\kappa = 4$). Ratios are means of cell-level ratios over $J \in \{25, 50, 100\}$, $\bar{n} \in \{20, 50, 100\}$, and the three effect distributions, with common random numbers across CV levels within each family.

Appendix G. Details of the Online Credit Recovery Study Analysis

G.1. Data and Site Definition. We use the public-use impact-analysis file of the Online Credit Recovery Study (American Institutes for Research, 2024), which contains the 1,737 student-by-subject records entering the study’s intent-to-treat analyses for the 2018 and 2019 summer terms (1,683 unique students; 54 appear in both subjects). Students who failed Algebra 1 or English 9 in ninth grade were randomized to an online credit-recovery class or the school’s teacher-directed class within randomization blocks defined by subject, summer cohort, and school, and in some schools also by the semester of course failure (Rickles et al., 2024). We define sites as school-by-subject cells, collapsing the two English 9 semester courses (E9A, E9B) to one English site per school because each student appears once per subject; this yields $J = 32$ sites (13 Algebra, 19 English), every one of which enrolled both arms, with site sizes 24 to 104 (median 49) and at least 8 students per arm in every site. The 32

sites contain 144 randomization blocks; 28 sites contain more than one block (up to 10), and 6 sites enrolled students in both summer terms.

G.2. Outcomes. The primary outcome is passing the credit-recovery course (`scm_passcrs`), which is recorded for every student; grades of D or better count as passing and incompletes as not passing, following the study’s convention. The distal outcome is subject-matched credits earned by the end of year four (`cby_y4_math` for Algebra sites, `cby_y4_eng` for English sites; observed ranges 0–90 and 0–87.5), also complete. The credits-earned measures used in the study’s own analyses cap subject-area credits at 80; the subject-specific versions used here are released without that cap. We do not use the end-of-course test score as a primary illustration because it is missing for 39% of students with differential attrition by arm; the diagnostics require credible sampling variances, which the administrative outcomes provide without modeling nonresponse.

G.3. Block-Adjusted Site-Level Estimation. Because randomization was blocked, the site-level first stage respects the blocks. Within each site, the intent-to-treat effect $\hat{\tau}_j$ is the treatment coefficient from an ordinary least squares regression of the outcome on the assignment indicator and randomization-block fixed effects, with HC2 heteroskedasticity-robust standard errors; blocks that contain a single arm within a site are absorbed by the fixed effects, as the design requires. The adjustment matters in principle because treated shares differ across blocks: in the six sites spanning both summer terms, the treated share differs between terms by up to .312 (median .081), so a raw difference in site means would mix term-to-term composition with the treatment contrast. In these data the adjustment is modest in practice: the block-adjusted $\hat{\tau}_j$ differs from the unblocked difference in means by at most .243 (median .029) on the risk-difference scale, and the pooled diagnostics move little.

The covariate-adjusted sensitivity analysis adds to each within-site regression the grade-8 state test score in the matching subject (mean-imputed, with its missingness flag) and ninth-grade GPA (imputed, with flag), taking the treatment coefficient and its HC2 standard error. Adjustment changes the pooled results little (Table 5) because randomization balances the covariates within blocks; its main effect is a modest reduction in the v_j .

G.4. Model Fitting and Diagnostics. The site pairs $(\hat{\tau}_j, v_j)$ enter the normal-normal model exactly as in the main text, fit by the grid method of Appendix D

with a uniform prior on (τ, σ) and $M = 20,000$ draws. All diagnostics are posterior functionals; the plug-in column of Table 5 evaluates the same functionals at the posterior mean of σ^2 , and the $\Pr(I > \frac{1}{2})$ column is the posterior probability that informativeness exceeds the halfway point of the reliability scale. Because the sampling variances are treated as known, the logit-scale profile variance V carries no posterior uncertainty (Appendix C); its values are .237 (passing, pooled), .158 (Algebra), .282 (English), .260 (covariate-adjusted), and .313 (credits), and the maximum-to-minimum sampling-variance ratios are 7.9, 8.9 (adjusted), and 10.4 (credits).

Two caveats apply. One is that the pooled analyses place both subjects in one profile; the subject-specific fits in Table 5 show the same qualitative story with the wider intervals that $J = 13$ or 19 sites imply. The other is that effects for the binary passing outcome are risk differences, and the normal first stage is an approximation whose quality is weakest in the smallest sites; the site-size floor of 24 students (8 per arm) keeps the approximation serviceable, and the covariate-adjusted analysis, which relies on the same approximation, reproduces the diagnostics nearly unchanged. The scale of the effect is itself a modeling choice, examined next.

G.5. Sensitivity Analyses. Three design features of these data could distort the diagnostics: cross-site dependence through schools that contribute two sites, the choice of effect scale for a binary outcome, and first-stage estimation error in $(\hat{\tau}_j, v_j)$ that the known-variance model ignores. Each is examined in turn.

Dependence through paired schools. Eight schools contribute both an Algebra and an English site, so the pooled profile is not a sample of independent sites. Refitting the passing analysis on all $2^8 = 256$ one-site-per-paired-school selections ($J = 24$ each) yields posterior-mean informativeness between .611 and .748 (median .690); the credits analysis yields .115 to .427 (median .263). No selection reverses either conclusion: the proximal I exceeds one half in every selection, and the distal I stays below it.

Effect scale. On the log-odds scale (Woolf variances; the 0.5 continuity correction was not needed in any site-by-arm cell), the passing profile is markedly less informative: $I = .530$ [.252, .745] with $\Pr(I > \frac{1}{2}) = .62$, $H = .035$, and $V = .306$, against .671 [.499, .813], .97, .025, and .237 on the risk-difference scale. The discrepancy is expected: risk differences near a common baseline compress the effect spread relative to log odds, so the same data can appear informative on one scale and equivocal on the other. Informativeness is a property of the profile on a stated scale, and reports

should state the scale; we adopt the risk difference as primary because the study’s own impact analysis is reported on that scale (Rickles et al., 2024).

First-stage uncertainty. A student-level bootstrap ($B = 500$), resampling within site-by-arm-by-block cells and recomputing the full pipeline (block-adjusted first stage, grid posterior, diagnostics) on each draw, propagates estimation error in $(\hat{\tau}_j, v_j)$. Resampled binary cells can degenerate (all passes or all failures), sending an HC2 variance to zero, so bootstrap sampling variances are floored: for the passing outcome at the Agresti-style continuity-corrected minimum $\sum_a \tilde{p}_a(1 - \tilde{p}_a)/n_a$ with $\tilde{p}_a = (x_a + 0.5)/(n_a + 1)$, and for credits at 5% of the site’s original v_j . The central 95% intervals are $I \in [.664, .828]$, $H \in [.015, .031]$, and $V \in [.191, .324]$ for passing, and $I \in [.377, .727]$, $H \in [.025, .052]$, and $V \in [.234, .468]$ for credits. The passing interval sits slightly above the posterior interval’s center because resampling adds first-stage noise to the spread of the site estimates, part of which the refitted model attributes to σ^2 ; the shift is toward the individual-inference conclusion rather than away from it. For credits the bootstrap interval is wide and overlaps one half, reinforcing the main text’s caution that the distal diagnostic supports only ensemble-level claims.

G.6. Replication. The script `04_ocrs.R` reproduces Table 5, the site-level estimates (`tbl_ocrs_sites.csv`), and all three sensitivity analyses (`tbl_ocrs_sensitivity.csv`) from the public data release, available from openICPSR project 202181 (American Institutes for Research, 2024); the figure scripts distributed with the replication materials reproduce Figure 8. The data path is supplied through the `OCRS_DATA` environment variable, the seed is fixed, and the full pipeline including the bootstrap runs in about two minutes on a 32-vCPU instance.